# Course Project


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Course Code: ELEC 8900-67: Power System Analysis & Control

Date: Mar. 30, 2026

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# Abstract

This project evaluates the performance of a modified IEEE 14-bus power system under steady-state, short-circuit, and voltage-stability conditions using PSAT and MATLAB. The system was modified by adding an extra line between Buses 6 and 11, reducing all transmission-line and two-winding transformer impedances by 5%, and reducing all generator and load powers by 5%. First, a power flow study was carried out to determine the base operating condition, assess the effect of generator reactive power limits, and compare shunt capacitor placement under a 45% load increase. Second, a short-circuit study was performed at Bus 6 using sequence-network analysis to calculate the fault currents and post-fault bus voltages for a three-phase solid fault and a single-line-to-ground solid fault. Third, continuation power flow was used to obtain the PV curve at Bus 11 for the base case and the Line 2-4 tripped case, followed by ATC calculation and N-1 assessment. The results show that the base system operates acceptably, that Bus 14 provides better shunt compensation than Bus 12 under stressed loading, that the three-phase fault is the most severe fault at Bus 6, and that the Line 2-4 outage reduces the voltage-stability margin and limits the N-1 secure ATC of the system.

# Introduction

The results obtained show that the base system is operating satisfactorily, that Bus 14 has better compensation compared to Bus 12 when subjected to a stress condition, that the three-phase fault is the most severe fault at Bus 6, and that the outages of Line 2-4 decrease the voltage stability margin and limit the N-1 secure ATC of the system. These analyses were used in this project in a modified IEEE 14-Bus system. The modifications were: an additional transmission line connecting buses 6 and 11, a 5% decrease in all lines and transformer impedances, and a 5% decrease in all generator and load powers. The modified system used in this project is shown in Figure 1. The grounding assumptions were particularly important in the short circuit analysis. All generators and synchronous condensers were delta-connected except for the two transformers at bus 4, and synchronous condenser at bus 8 which are Y-grounded. The transformer connections at bus 4 were modeled based on their grounding and winding configurations in the zero-sequence network. All outputted scripts are included in the appendix.

# Nomenclature

- $Y_{Bus}$: Bus admittance matrix
- $Y_1$: Positive-sequence bus admittance matrix
- $Y_2$: Negative-sequence bus admittance matrix
- $Y_0$: Zero-sequence bus admittance matrix
- $Z_{BUS}$: Bus impedance matrix
- $Z_1$: Positive-sequence Thevenin impedance at the faulted bus
- $Z_2$: Negative-sequence Thevenin impedance at the faulted bus
- $Z_0$: Zero-sequence Thevenin impedance at the faulted bus
- $V$: Bus voltage magnitude
- $V_{11}$: Voltage magnitude at Bus 11
- $V_{14}$: Voltage magnitude at Bus 14
- $V_{12}$: Voltage magnitude at Bus 12
- $V_{prefault}$: Pre-fault bus voltage
- $V_{min}$: Minimum allowable bus voltage
- $V_{max}$: Maximum allowable bus voltage
- $P$: Active power
- $Q$: Reactive power
- $P_L$: Active load power
- $Q_L$: Reactive load power
- $P_G$: Generated active power
- $Q_G$: Generated reactive power
- $\lambda$: Loading parameter in continuation power flow
- $\lambda_{initial}$: Initial loading parameter used in CPF
- $\lambda_{crit}$: Critical loading parameter at the selected voltage limit
- $\lambda_{max}$: Maximum loading parameter at the nose point
- TTC : Total Transfer Capability
- ETC : Existing Transmission Commitment
- TRM : Transmission Reliability Margin

- ATC : Available Transfer Capability
- CPF : Continuation Power Flow
- PV curve : Plot of bus voltage magnitude versus loading parameter
- N-1 : Security criterion requiring the system to remain acceptable after the loss of one component
- p.u. : Per unit
- MW : Megawatt
- Mvar : Megavolt-ampere reactive
- kV : Kilovolt
- kA : Kiloampere
- $a$: Off-nominal transformer tap ratio

# Discussion

In this report, PSAT was used for power flow and continuation power flow (CPF), MATLAB was used for sequence $Y_{BUS}$ Construction and fault calculations, Q1(a) was used as the pre-fault operating point for Q2, and lastly the setting that was used in PSAT for power flow and CPF are:
General Settings:

- Discard Dynamic Comp
- Use Distributed Slack Bus
- Check PV reactive limits

CPF Settings:

- Qg limit controls
- Initial Loading Param. = 0.99
- Step Size Control = 0.1

These settings were used consistently throughout the project to ensure that the power flow, continuation power flow (CPF), and voltage-stability results were obtained under the same modeling assumptions.
And the project involving solving the following 14-Bus system shown in figure 1:

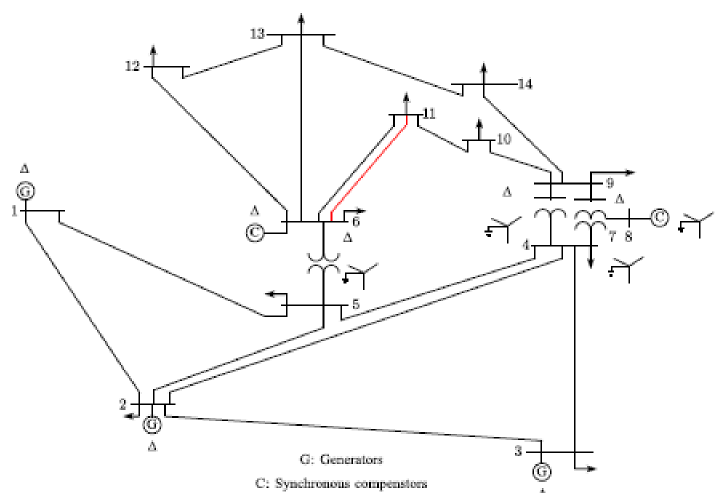

*Figure 1: 14-bus IEEE benchmark system*

Figure 1 shows the modified IEEE 14-bus benchmark system with adding the new line 6-11 and that was used throughout the project for steady-state, short-circuit, and voltage-stability analysis.

## Question 1

In this question, the steady-state performance of the modified IEEE 14-bus system using power flow was analyzed. The system was modified by adding an extra transmission line between Buses 6 and 11, reducing all line and transformer impedances by 5%, and reducing all generator and load powers by 5%. The purpose of this question is to determine the base operating condition, compare the solution with and without generator reactive power limits, and examine shunt capacitor placement under a 45% load increase.

### a) Determine the base power flow solution

The base-case power flow converged successfully, which shows that the modified IEEE 14-bus system is numerically stable under the specified operating condition. All bus voltages remained within limits, the real and reactive power balance was satisfied, and the system operated acceptably after the network modifications. The main base-case power flow busses results are summarized in Table 1:

| POWER FLOW RESULTS | | | | | | |
|---|---|---|---|---|---|---|
| Bus | V | phase | P gen | Q gen | P load | Q load |
| | [p.u.] | [rad] | [p.u.] | [p.u.] | [p.u.] | [p.u.] |
| Bus 01 | 1.06 | 0 | 2.2601 | -0.11467 | 0 | 0 |
| Bus 02 | 1.0324 | -0.18264 | 0.38 | 0.5 | 0.20615 | 0.12065 |
| Bus 03 | 1.01 | -0.29708 | 0 | 0.32678 | 0.8949 | 0.1805 |
| Bus 04 | 1.0092 | -0.24536 | 0 | 0 | 0.4541 | 0.038 |
| Bus 05 | 1.0114 | -0.2147 | 0 | 0 | 0.0722 | 0.0152 |
| Bus 06 | 1.0658 | -0.31052 | 0 | 0.24 | 0.1064 | 0.07125 |
| Bus 07 | 1.0509 | -0.29077 | 0 | 0 | 0 | 0 |
| Bus 08 | 1.0897 | -0.29077 | 0 | 0.24 | 0 | 0 |
| Bus 09 | 1.0375 | -0.31453 | 0 | 0 | 0.28025 | 0.1577 |
| Bus 10 | 1.0458 | -0.31799 | 0 | 0 | 0.0855 | 0.0551 |
| Bus 11 | 1.0538 | -0.31505 | 0 | 0 | 0.03325 | 0.0171 |
| Bus 12 | 1.0484 | -0.32596 | 0 | 0 | 0.05795 | 0.0152 |
| Bus 13 | 1.0423 | -0.32615 | 0 | 0 | 0.12825 | 0.0551 |
| Bus 14 | 1.0265 | -0.33245 | 0 | 0 | 0.14155 | 0.0475 |

*Table 1: Power flow results of 14-Bus system*

A key observation is that generators at Buses 2, 6, and 8 reached their maximum reactive power limits, as shown in Table 1. This means that these generators were no longer able to supply additional reactive power to maintain their scheduled voltages, so the actual voltages at those buses fell slightly below their set points. This indicates that the system is already relying on significant reactive support under normal operating conditions and that the reactive reserve margin is limited.

### b) Comparing results

With generator reactive power limits removed, the power-flow solution converged in four iterations, and the voltage profile improved slightly, as shown in Table 2. The generator buses were able to remain closer to their scheduled voltages because the reactive power limits no longer restricted their output. For example, Bus 2 held 1.045 p.u., Bus 6 held 1.070 p.u., and Bus 8 held 1.090 p.u. However, this improvement came at the expense of unrealistic reactive power generation. In particular, Bus 2 supplied

0.73708 p.u. reactive power, which exceeds its original upper limit of 0.50 p.u. Therefore, the case without Q-limits is useful for comparison, but the case with Q-limits is the more realistic operating condition.

| **Quantity** | **Q1(a), With Q-limits** | **Q1(b), Without Q-limits** |
|---|---|---|
| Number of iterations | 8 | 4 |
| Bus 2 voltage, p.u. | 1.0324 | 1.0450 |
| Bus 2 reactive power, p.u. | 0.50000 | 0.73708 |
| Bus 3 reactive power, p.u. | 0.32678 | 0.22125 |
| Bus 6 voltage, p.u. | 1.0658 | 1.0700 |
| Bus 6 reactive power, p.u. | 0.24000 | 0.23177 |
| Bus 8 voltage, p.u. | 1.0897 | 1.0900 |
| Bus 8 reactive power, p.u. | 0.24000 | 0.22114 |
| Bus 11 voltage, p.u. | 1.0538 | 1.0580 |
| Bus 14 voltage, p.u. | 1.0265 | 1.0307 |
| Total real losses ($P_L$), p.u. | 0.17964 | 0.18103 |
| Total reactive losses ($Q_L$), p.u. | 0.41880 | 0.41659 |

*Table 2: Comparing results with and without generator Q limits*

As shown in Table 2, removing the generator reactive power limits improves the voltage profile slightly, but this occurs because some generators, especially Bus 2, supply reactive power beyond their practical operating limits.

## c) 45% load increase and shunt capacitor placement at Buses 12 and 14

With generator Q-limits in place, the active and reactive power demands were increased by 45%. Shunt compensation was performed by modeling Bus 12 and Bus 14 as separate PV buses regulated at 1.0 p.u. In the case of Bus 12, the voltage at Bus 12 was regulated to 1.0 p.u., while the required amount of reactive injection was 0.35815 p.u., which translates into a value of 35.815 Mvar on a 100 MVA base. As can be seen in Table 3, only two voltage violations remained, at Bus 3 and Bus 4, which means that the compensation effort of Bus 12 was essentially local. Since PSAT was not forced to use constant PQ loads, the total loads listed in Table 3 correspond to the solved operating point and hence are slightly different from the nominal scheduled ones.

| **Summary Table for Bus 12 case** | |
|---|---|
| **Quantity** | **Value** |
| Regulated bus | Bus 12 |
| Target voltage, p.u. | 1.0000 |

| Summary Table for Bus 12 case | |
|---|---|
| **Quantity** | **Value** |
| Required reactive injection, p.u. | 0.35815 |
| Required reactive injection, Mvar | 35.815 |
| Bus 3 voltage, p.u. | 0.88256 |
| Bus 4 voltage, p.u. | 0.89389 |
| Number of voltage violations | 2 |
| Total solved real load, p.u. | 3.5090 |
| Total solved reactive load, p.u. | 1.1105 |

*Table 3: Shunt capacitor results for Bus 12 placement*

For the Bus 14 case, the voltage at Bus 14 was maintained at 1.0 p.u. and the reactive injection required was increased to 0.47430 p.u. or 47.430 Mvar. From the results presented in Table 4, it was observed that the voltage violations were reduced to only one at Bus 3. The voltage levels at the downstream buses, especially Buses 9 to 13, were higher than those in the Bus 12 case. Therefore, although the capacitor required at Bus 14 was larger, the voltage profile was better. The results presented in Table 5 confirm this observation because the Bus 14 placement not only reduced the number of voltage violations from two to one but provided stronger voltage support.

| Summary Table for Bus 14 Case | |
|---|---|
| **Quantity** | **Value** |
| Regulated bus | Bus 14 |
| Target voltage, p.u. | 1.0000 |
| Required reactive injection, p.u. | 0.47430 |
| Required reactive injection, Mvar | 47.430 |
| Bus 3 voltage, p.u. | 0.89415 |
| Number of voltage violations | 1 |
| Total solved real load, p.u. | 3.5509 |
| Total solved reactive load, p.u. | 1.1179 |

*Table 4: Shunt capacitor results for Bus 14 placement*

The overall comparison of the results is presented in Table 5. Bus 14 is the better choice for capacitor placement because not only is the voltage support better, but the voltage violations are reduced from two to one, even though the capacitor size is larger.

| Summary Comparison Table for Shunt Capacitor Placement | | |
|---|---|---|
| **Quantity** | **Bus 12 placement** | **Bus 14 placement** |
| Required reactive injection, p.u. | 0.35815 | 0.47430 |

| Summary Comparison Table for Shunt Capacitor Placement | | |
|---|---|---|
| **Quantity** | **Bus 12 placement** | **Bus 14 placement** |
| Required reactive injection, Mvar | 35.815 | 47.430 |
| Regulated bus voltage, p.u. | 1.0000 | 1.0000 |
| Bus 3 voltage, p.u. | 0.88256 | 0.89415 |
| Bus 4 voltage, p.u. | 0.89389 | 0.90990 |
| Bus 9 voltage, p.u. | 0.91671 | 0.96148 |
| Bus 10 voltage, p.u. | 0.93568 | 0.96173 |
| Bus 11 voltage, p.u. | 0.95031 | 0.97029 |
| Bus 12 voltage, p.u. | 1.0000 | 0.96957 |
| Bus 13 voltage, p.u. | 0.94962 | 0.97086 |
| Bus 14 voltage, p.u. | 0.90952 | 1.0000 |
| Number of voltage violations | 2 | 1 |
| **Better placement** | | **Bus 14** |

*Table 5: Comparison of shunt capacitor placement at Bus 12 and Bus 14*

**Shunt capacitor calculation**
The capacitor size is taken directly from the reactive power injected by the compensated PV bus. The formula is:

$$Q_c \text{ (Mvar)} = Q_{inj} \text{ (p.u.)} \times S_{base} \text{ (MVA)}$$

Since system base is:

$$S_{base} = 100 \text{ MVA}$$

**Bus 12**
From PSAT:

$$Q_{inj,12} = 0.35815 \text{ p.u.}$$

Therefore,

$$Q_{c,12} = 0.35815 \times 100 = 35.815 \text{ Mvar}$$

**Bus 14**
From PSAT:

$$Q_{inj,14} = 0.47430 \text{ p.u.}$$

Therefore,

$$Q_{c,14} = 0.47430 \times 100 = 47.430 \text{ Mvar}$$

Although a larger capacitor is required at Bus 14 than at Bus 12, a better voltage profile is achieved with fewer voltage violations reduced from two to one. On the other hand, although the Bus 12 placement results in a more economical reactive size, a weaker voltage response results. Hence, from the voltage support and security point of view, the better capacitor placement for the 45% increase in loading condition is at bus 14. This means that the Bus 14 location provides stronger system-wide voltage support under the stressed loading condition.

# Question 2

## a) Construct $Y_{BUS}$ Matrices

In the short circuit analysis, the pre-fault voltage values obtained in Question 1(a) were used. Using the modified data and the grounding scheme assumed in the project, the positive, negative, and zero sequence $Y_{BUS}$ matrices were generated using MATLAB code. The positive sequence matrix was directly generated using the modified data for the lines and the transformers, including the series admittance, the shunt charging susceptance, and the off-nominal transformer ratio 'a' the negative sequence network was assumed to be the same as the positive sequence network, and thus Y2 = Y1. The zero-sequence network is constructed independently based on the grounding assumptions. The generators and synchronous condensers are delta-connected except for the condenser connected to Bus 8, which is Y-connected to ground. The loads are all Y-connected to ground. The transformer between Buses 5 and 6 is represented by a grounding reactance of j0.1 p.u. In addition, the transformer between Buses 4 and 9 is represented by a grounded-wye/delta connection so that it provides a zero-sequence grounding path to the Bus 4 side but blocks zero sequence current flow to the Bus 9 side. The sequence network modeling assumptions are summarized in Table 6, and the sequence $Y_{BUS}$ matrices are provided in the appendix.

| **Modeling Assumptions Summary Table** | |
|---|---|
| **Item** | **Modeling assumption** |
| Pre-fault operating point | Obtained from Q1(a) power flow results |
| Positive-sequence network | Built from modified line and transformer data |
| Negative-sequence network | Assumed identical to positive-sequence network, ($Y_2 = Y_1$) |
| Zero-sequence network | Built separately using grounding and transformer connection assumptions |
| Generator grounding | All generators and synchronous condensers delta-connected except Bus 4, which is Y-grounded |
| Load grounding | All loads assumed Y-grounded |
| Transformer 5-6 grounding | Grounded through reactance (j0.1) p.u. |
| Transformer modeling | Off-nominal transformer tap ratio (a) included in $Y_{BUS}$ formulation |

*Table 6: Summary of sequence-network modeling assumptions used in Question 2*

## b) Three-phase fault and single-line-to-ground fault at Bus 6

With the sequence networks obtained in part (a), the short-circuit analysis was carried out for Bus 6 for two types of faults: three-phase solid faults and single-line-to-ground solid faults. The Thevenin equivalent impedances referred to the faulted buses were obtained as follows:

$$\mathbf{Z_1=Z_2=0.013710+j0.110986\ p.u.\ \&\ Z_0=1.144915+j0.732181\ p.u.}$$

These values are summarized in Table 7. The much larger zero-sequence impedance is consistent with the grounding structure of the system. For the three-phase solid fault at Bus 6, the fault current was 39.873 kA, as shown in Table 7

| **Thevenin impedances and fault currents** | |
|---|---|
| **Quantity** | **Value** |
| ($Z_1$), p.u. | (0.013710 + j0.110986) |
| ($Z_2$), p.u. | (0.013710 + j0.110986) |
| ($Z_0$), p.u. | (1.144915 + j0.732181) |
| Three-phase fault current, kA | 39.873 |
| Single-line-to-ground fault current, kA | 8.850 |

*Table 7: Thevenin equivalent impedances and calculated fault currents at Bus 6*

. This is the highest fault current because the three-phase fault involves only the positive-sequence network. The voltage at the faulted bus collapsed to zero, and large voltage depressions appeared at other buses. Selected post-fault voltages are shown in Table 8 (full report short circuit report is shown in the appendix).

| **Bus** | **Three-phase fault, L-G (A,B,C) voltage (kV)** | **SLG fault phase A, L-G voltage (kV)** | **SLG fault phase B, L-G voltage (kV)** | **SLG fault phase C, L-G voltage (kV)** |
|---|---|---|---|---|
| 6 | 0.0000 | 0.0000 | 12.8857 | 14.2978 |
| 7 | 5.0683 | 8.0361 | 8.1366 | 8.4407 |
| 10 | 1.4568 | 1.0801 | 11.8813 | 13.6868 |
| 11 | 0.7906 | 0.6041 | 12.3099 | 13.9285 |
| 12 | 0.3605 | 0.5122 | 12.2446 | 13.9332 |
| 14 | 2.3710 | 1.8860 | 11.0444 | 13.1872 |

*Table 8: Selected bus voltages during three-phase and single-line-to-ground faults at Bus 6*

For the single-line-to-ground solid fault at Bus 6, the fault current was 8.850 kA, which is much lower than the three-phase fault current because the SLG fault includes the positive-, negative-, and zero-sequence impedances, and the zero-sequence impedance is large. At Bus 6, phase A collapsed to 0 kV, while phases B and C remained at 12.8857 kV and 14.2978 kV, respectively, which is the expected unbalanced fault behaviour.
Therefore, the worst fault at Bus 6 is the three-phase solid fault because it produces the highest fault current and imposes the highest electrical duty on the system. The single-line-to-ground fault is less severe because it is limited by the high zero-sequence impedance introduced by the system grounding and transformer connections.

## Question 3

### a) Interpret the PV curve at Bus 11 for the base and the Line 2-4 tripped system

The PV curves for Bus 11 in the base case and the Line 2-4 tripped case are shown in Figure 2.

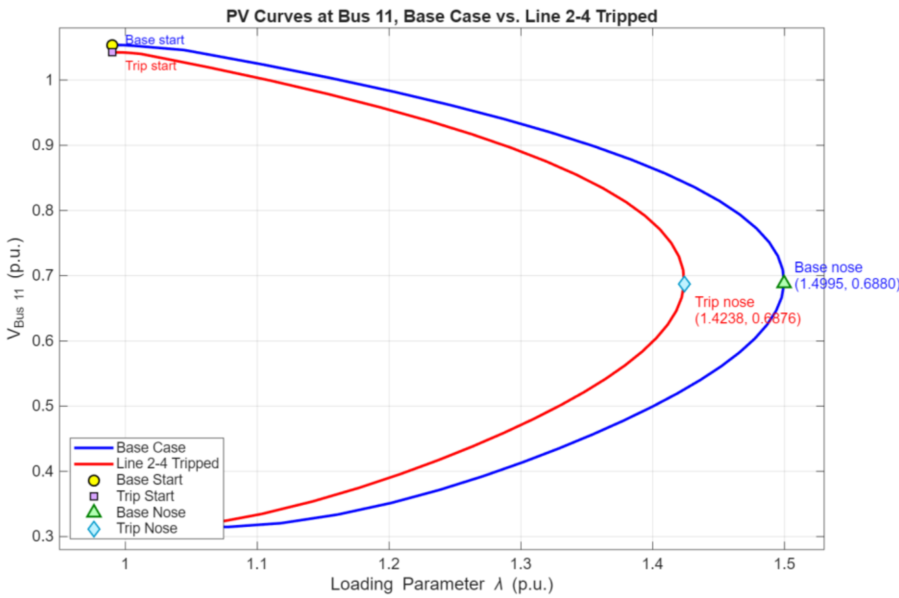


*Figure 2: PV curves of Bus 11 for the base case and the Line 2-4 tripped case*

In the base case, the PV curve has shown the normal nose curve behavior. The voltage at Bus 11 has decreased gradually with an increase in the loading parameter until a point where the decrease was more pronounced. The point at which maximum loading was achieved occurred at $\lambda_{base} = 1.49955$, at which point the voltage at Bus 11 was 0.68798 p.u. and the total real load was 368.97 MW. The principal base case PV curve values are shown in Table 9.

| **Quantity** | **Value** |
|---|---|
| Maximum loading parameter, $\lambda_{max}$ | 1.49955 |
| Bus 11 voltage at nose point, p.u. | 0.68798 |
| Total real load at nose point, MW | 368.97 |
| Critical loading parameter at ($V_{11}$=0.9) p.u. | 1.3531 |

*Table 9: Base-case Bus 11 PV-curve critical values*

The critical loading parameter was not determined at the nose point, but where the PV curves intersected with the voltage criterion at 0.9 p.u. on the voltage axis. By using linear interpolation, we determined that $\lambda_{crit,base}\approx 1.3531$. Following the course method demonstrated in class, the critical loading parameter for ATC calculation was taken from the intersection of the Bus 11 PV curve with the $V = 0.9$p.u. criterion.

**$\lambda_{crit,base}$ calculation:**

From Bus 11 base PV table, the two points around $V = 0.9$ were:

- $(\lambda, V_{11}) = (1.32245, 0.91979)$

- $(\lambda, V_{11}) = (1.35494, 0.89883)$

Since 0.9 lies between 0.91979and 0.89883, we used linear interpolation:

$$\lambda_{crit,base} = 1.32245 + \frac{0.91979 - 0.9}{0.91979 - 0.89883}(1.35494 - 1.32245)$$
$$\lambda_{crit,base} \approx 1.3531$$

This indicates that the operating limit is attained before voltage collapse.

For the tripped case, the PV curve for Bus 11 shifted downwards and to the left, as shown above in Figure 2, indicating a weakening in the system due to the outage of Line 2-4, which resulted in a reduction in voltage stability margin. The maximum loading point for this case was determined at a value of λ=1.42377, with a voltage level of 0.68758 p.u. at Bus 11, and a total real power demand of 350.32 MW. The key PV curve parameters for the tripped case are presented in Table 10 below (Bus 11 PV data for the tripped case are also included in the appendix).

| **Tripped Case Summary Table** | |
|---|---|
| **Quantity** | **Value** |
| Maximum loading parameter, $\lambda_{max}$ | 1.42377 |
| Bus 11 voltage at nose point, p.u. | 0.68758 |
| Total real load at nose point, MW | 350.32 |
| Critical loading parameter at ($V_{11}$=0.9) p.u. | 1.2870 |

*Table 10: Tripped-case Bus 11 PV-curve critical values*

The line flows were zero on Line 2-4 in the contingency case, indicating a correct modeling of the line outage. By using a similar value of 0.9 p.u. for voltage level at Bus 11 and carrying out a linear interpolation, a value of λcrit,trip ≈1.2870 was found for the tripped case.
**λcrit, trip calculation**:

From tripped Bus 11 PV table, the two points around $V = 0.9$ were:

- $(\lambda, V_{11}) = (1.26265, 0.91699)$
- $(\lambda, V_{11}) = (1.29245, 0.89623)$

So again, by linear interpolation:

$$\lambda_{crit,trip} = 1.26265 + \frac{0.91699 - 0.9}{0.91699 - 0.89623}(1.29245 - 1.26265)$$
$$\lambda_{crit,trip} \approx 1.2870$$

Since this value is lower than the base-case value, the outage of Line 2-4 reduces the allowable transfer margin and becomes the limiting contingency.

## b) Determine ATC and assess the N-1 criterion

Following the course method demonstrated in class, the critical loading parameter for ATC calculation was taken at the intersection of the Bus 11 PV curve with the 0.9 p.u. voltage criterion.

ATC calculation:

$$ATC = TTC - ETC - TRM.$$

where $TTC$ is the total transfer capability, $ETC$ is the existing transmission commitment, and $TRM$ is the transmission reliability margin. The reliability margin was taken as

$$TRM = 0.05 * TTC.$$

The base operating real load from Question 1(a) was

$$P_{base,total} = 2.4605 \text{ p.u.} = 246.05 \text{ MW}.$$

Using the CPF initial loading parameter of $\lambda_{initial} = 0.99$, the existing transmission commitment becomes

$$ETC = 246.05 \times 0.99 = 243.59 \text{ MW}$$

**Base case**

Using $\lambda_{crit,base} = 1.3531$,

$$TTC_{base} = 246.05 \times 1.3531 = 332.94 \text{ MW}$$
$$TRM_{base} = 0.05 \times 332.94 = 16.65 \text{ MW}$$
$$ATC_{base} = 332.94 - 243.59 - 16.65 = 72.70 \text{ MW}$$

**Tripped case**

Using $\lambda_{crit,trip} = 1.2870$,

$$TTC_{trip} = 246.05 \times 1.2870 = 316.68 \text{ MW}$$
$$TRM_{trip} = 0.05 \times 316.68 = 15.83 \text{ MW}$$
$$ATC_{trip} = 316.68 - 243.59 - 15.83 = 57.25 \text{ MW}$$

These results show that the contingency reduces ATC from **72.70 MW** in the base case to **57.25 MW** in the tripped case. These results are presented in Table 11.

| **ATC Calculations Summary** | | |
|---|---|---|
| **Quantity** | **Base case** | **Tripped case** |
| $\lambda_{crit}$ at ($V_{11}$=0.9) p.u. | 1.3531 | 1.2870 |
| TTC, MW | 332.94 | 316.68 |
| ETC, MW | 243.59 | 243.59 |
| TRM, MW | 16.65 | 15.83 |
| ATC, MW | 72.70 | 57.25 |

*Table 11: ATC calculation summary for the base case and tripped case*

The result for the Line 2-4 tripped contingency has the lower ATC value and hence is the determining contingency, and its result governs the N-1 evaluation. Thus, the N-1 secure transfer capability of the system has been determined to be 57.25 MW, based on the adopted criterion of V11=0.9 p.u. Any plan exceeding this additional transfer capability would violate the N-1 security criterion. The decrease in λcrit and ATC values also confirms that the outage of line 2-4 has weakened the system and reduced its transfer capability.

# Conclusion

This project provided a complete evaluation of the modified IEEE 14-bus system under steady-state, short-circuit, and voltage-stability conditions. The power flow study showed that the system operates acceptably in the base case, but several generators are already at their Q-limits, which means that the reactive reserve is limited. Under stressed loading, Bus 14 provided better shunt compensation than Bus 12, although it required a larger capacitor.

The short-circuit study showed that the three-phase solid fault at Bus 6 is the most severe fault because it produces the highest fault current, while the single-line-to-ground fault is limited by the large zero-sequence impedance created by the system grounding and transformer configuration.

The voltage-stability study showed that the Line 2-4 outage weakens the system, shifts the Bus 11 PV curve downward and to the left, and reduces the available transfer capability from 72.70 MW to 57.25 MW. Therefore, the Line 2-4 tripped case is the limiting contingency, and the N-1 secure ATC of the modified IEEE 14-bus system is 57.25 MW.

# Appendix

## Appendix A power flow Q1(a)

NETWORK STATISTICS

| | |
|---|---|
| Buses: | 14 |
| Lines: | 21 |
| Generators: | 5 |
| Loads: | 11 |

SOLUTION STATISTICS

| | |
|---|---|
| Number of Iterations: | 8 |
| Maximum P mismatch [p.u.] | 0 |
| Maximum Q mismatch [p.u.] | 0 |
| Power rate [MVA] | 100 |

POWER FLOW RESULTS

| Bus | V [p.u.] | phase [rad] | P gen [p.u.] | Q gen [p.u.] | P load [p.u.] | Q load [p.u.] |
|---|---|---|---|---|---|---|
| Bus 01 | 1.06 | 0 | 2.2601 | -0.11467 | 0 | 0 |
| Bus 02 | 1.0324 | -0.18264 | 0.38 | 0.5 | 0.20615 | 0.12065 |
| Bus 03 | 1.01 | -0.29708 | 0 | 0.32678 | 0.8949 | 0.1805 |
| Bus 04 | 1.0092 | -0.24536 | 0 | 0 | 0.4541 | 0.038 |
| Bus 05 | 1.0114 | -0.2147 | 0 | 0 | 0.0722 | 0.0152 |
| Bus 06 | 1.0658 | -0.31052 | 0 | 0.24 | 0.1064 | 0.07125 |
| Bus 07 | 1.0509 | -0.29077 | 0 | 0 | 0 | 0 |
| Bus 08 | 1.0897 | -0.29077 | 0 | 0.24 | 0 | 0 |
| Bus 09 | 1.0375 | -0.31453 | 0 | 0 | 0.28025 | 0.1577 |
| Bus 10 | 1.0458 | -0.31799 | 0 | 0 | 0.0855 | 0.0551 |
| Bus 11 | 1.0538 | -0.31505 | 0 | 0 | 0.03325 | 0.0171 |
| Bus 12 | 1.0484 | -0.32596 | 0 | 0 | 0.05795 | 0.0152 |
| Bus 13 | 1.0423 | -0.32615 | 0 | 0 | 0.12825 | 0.0551 |
| Bus 14 | 1.0265 | -0.33245 | 0 | 0 | 0.14155 | 0.0475 |

Maximum reactive power at bus <Bus 02>
Maximum reactive power at bus <Bus 06>
Maximum reactive power at bus <Bus 08>

LINE FLOWS

| From Bus | To Bus | Line | P Flow [p.u.] | Q Flow [p.u.] | P Loss [p.u.] | Q Loss [p.u.] |
|---|---|---|---|---|---|---|
| Bus 02 | Bus 05 | 1 | 0.22279 | 0.04429 | 0.00271 | -0.02546 |
| Bus 06 | Bus 12 | 2 | 0.07305 | 0.02892 | 0.00076 | 0.00158 |
| Bus 12 | Bus 13 | 3 | 0.01434 | 0.01213 | 8e-05 | 7e-05 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Bus 06 | Bus 13 | 4 | 0.16136 | 0.08744 | 0.00224 | 0.0044 |
| Bus 06 | Bus 11 | 5 | 0.06026 | 0.0556 | 0.00043 | 0.00089 |
| Bus 11 | Bus 06 | 6 | -0.06068 | -0.06032 | 0.00041 | 0.00096 |
| Bus 11 | Bus 10 | 7 | 0.08726 | 0.09793 | 0.00037 | 0.00099 |
| Bus 09 | Bus 10 | 8 | -0.00123 | -0.04151 | 0.00015 | 0.00033 |
| Bus 09 | Bus 14 | 9 | 0.09772 | 0.01024 | 0.00087 | 0.00184 |
| Bus 14 | Bus 13 | 10 | -0.0447 | -0.03911 | 0.00043 | 0.00089 |
| Bus 01 | Bus 02 | 11 | 1.1597 | -0.18096 | 0.07769 | 0.16231 |
| Bus 03 | Bus 02 | 12 | -0.61828 | 0.04133 | 0.0169 | 0.0278 |
| Bus 03 | Bus 04 | 13 | -0.27662 | 0.10495 | 0.0057 | -0.01896 |
| Bus 01 | Bus 05 | 14 | 1.1005 | 0.06628 | 0.05571 | 0.17982 |
| Bus 05 | Bus 04 | 15 | 0.73045 | -0.16938 | 0.00694 | 0.00949 |
| Bus 02 | Bus 04 | 16 | 0.39788 | 0.00533 | 0.00823 | -0.01206 |
| Bus 05 | Bus 06 | 17 | 0.46216 | 0.11039 | 0 | 0.0459 |
| Bus 04 | Bus 09 | 18 | 0.14133 | 0.01275 | 0 | 0.00981 |
| Bus 04 | Bus 07 | 19 | 0.2354 | -0.08833 | 0 | 0.01242 |
| Bus 08 | Bus 07 | 20 | 0 | 0.24 | 0 | 0.00854 |
| Bus 07 | Bus 09 | 21 | 0.2354 | 0.13071 | 0 | 0.00722 |

LINE FLOWS

| From Bus | To Bus | Line | P Flow [p.u.] | Q Flow [p.u.] | P Loss [p.u.] | Q Loss [p.u.] |
|---|---|---|---|---|---|---|
| Bus 05 | Bus 02 | 1 | -0.22007 | -0.06975 | 0.00271 | -0.02546 |
| Bus 12 | Bus 06 | 2 | -0.07229 | -0.02733 | 0.00076 | 0.00158 |
| Bus 13 | Bus 12 | 3 | -0.01426 | -0.01206 | 8e-05 | 7e-05 |
| Bus 13 | Bus 06 | 4 | -0.15912 | -0.08303 | 0.00224 | 0.0044 |
| Bus 11 | Bus 06 | 5 | -0.05983 | -0.05471 | 0.00043 | 0.00089 |
| Bus 06 | Bus 11 | 6 | 0.06109 | 0.06129 | 0.00041 | 0.00096 |
| Bus 10 | Bus 11 | 7 | -0.08689 | -0.09694 | 0.00037 | 0.00099 |
| Bus 10 | Bus 09 | 8 | 0.00139 | 0.04184 | 0.00015 | 0.00033 |
| Bus 14 | Bus 09 | 9 | -0.09685 | -0.00839 | 0.00087 | 0.00184 |
| Bus 13 | Bus 14 | 10 | 0.04513 | 0.03999 | 0.00043 | 0.00089 |
| Bus 02 | Bus 01 | 11 | -1.082 | 0.34326 | 0.07769 | 0.16231 |
| Bus 02 | Bus 03 | 12 | 0.63517 | -0.01353 | 0.0169 | 0.0278 |
| Bus 04 | Bus 03 | 13 | 0.28232 | -0.1239 | 0.0057 | -0.01896 |
| Bus 05 | Bus 01 | 14 | -1.0447 | 0.11353 | 0.05571 | 0.17982 |
| Bus 04 | Bus 05 | 15 | -0.72351 | 0.17887 | 0.00694 | 0.00949 |
| Bus 04 | Bus 02 | 16 | -0.38965 | -0.01739 | 0.00823 | -0.01206 |
| Bus 06 | Bus 05 | 17 | -0.46216 | -0.06449 | 0 | 0.0459 |
| Bus 09 | Bus 04 | 18 | -0.14133 | -0.00294 | 0 | 0.00981 |
| Bus 07 | Bus 04 | 19 | -0.2354 | 0.10075 | 0 | 0.01242 |
| Bus 07 | Bus 08 | 20 | 0 | -0.23146 | 0 | 0.00854 |
| Bus 09 | Bus 07 | 21 | -0.2354 | -0.12348 | 0 | 0.00722 |

GLOBAL SUMMARY REPORT

TOTAL GENERATION

REAL POWER [p.u.] 2.6401

REACTIVE POWER [p.u.] 1.1921

TOTAL LOAD

REAL POWER [p.u.] 2.4605
REACTIVE POWER [p.u.] 0.7733

TOTAL LOSSES

REAL POWER [p.u.] 0.17964
REACTIVE POWER [p.u.] 0.4188

LIMIT VIOLATION STATISTICS
ALL VOLTAGES WITHIN LIMITS.
ALL REACTIVE POWERS WITHIN LIMITS (3 BINDING).
ALL CURRENT FLOWS WITHIN LIMITS.
ALL REAL POWER FLOWS WITHIN LIMITS.
ALL APPARENT POWER FLOWS WITHIN LIMITS.

## Appendix B power flow Q1(b)

NETWORK STATISTICS

Buses: 14
Lines: 21
Generators: 5
Loads: 11

SOLUTION STATISTICS

Number of Iterations: 4
Maximum P mismatch [p.u.] 0
Maximum Q mismatch [p.u.] 0
Power rate [MVA] 100

POWER FLOW RESULTS

| Bus | V [p.u.] | phase [rad] | P gen [p.u.] | Q gen [p.u.] | P load [p.u.] | Q load [p.u.] |
|---|---|---|---|---|---|---|
| Bus 01 | 1.06 | 0 | 2.2615 | -0.22136 | 0 | 0 |
| Bus 02 | 1.045 | -0.18527 | 0.38 | 0.73708 | 0.20615 | 0.12065 |
| Bus 03 | 1.01 | -0.29476 | 0 | 0.22125 | 0.8949 | 0.1805 |
| Bus 04 | 1.0149 | -0.24499 | 0 | 0 | 0.4541 | 0.038 |
| Bus 05 | 1.0173 | -0.21473 | 0 | 0 | 0.0722 | 0.0152 |
| Bus 06 | 1.07 | -0.30965 | 0 | 0.23177 | 0.1064 | 0.07125 |
| Bus 07 | 1.0543 | -0.28996 | 0 | 0 | 0 | 0 |
| Bus 08 | 1.09 | -0.28996 | 0 | 0.22114 | 0 | 0 |
| Bus 09 | 1.0415 | -0.31352 | 0 | 0 | 0.28025 | 0.1577 |
| Bus 10 | 1.05 | -0.31703 | 0 | 0 | 0.0855 | 0.0551 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Bus 11 | 1.058 | -0.31413 | 0 | 0 | 0.03325 | 0.0171 |
| Bus 12 | 1.0526 | -0.32496 | 0 | 0 | 0.05795 | 0.0152 |
| Bus 13 | 1.0466 | -0.32514 | 0 | 0 | 0.12825 | 0.0551 |
| Bus 14 | 1.0307 | -0.33134 | 0 | 0 | 0.14155 | 0.0475 |

LINE FLOWS

| From Bus | To Bus | Line | P Flow [p.u.] | Q Flow [p.u.] | P Loss [p.u.] | Q Loss [p.u.] |
|---|---|---|---|---|---|---|
| Bus 02 | Bus 05 | 1 | 0.22386 | 0.08696 | 0.00302 | -0.02512 |
| Bus 06 | Bus 12 | 2 | 0.07308 | 0.02902 | 0.00076 | 0.00158 |
| Bus 12 | Bus 13 | 3 | 0.01438 | 0.01225 | 8e-05 | 7e-05 |
| Bus 06 | Bus 13 | 4 | 0.1614 | 0.08789 | 0.00222 | 0.00438 |
| Bus 06 | Bus 11 | 5 | 0.06033 | 0.05611 | 0.00043 | 0.0009 |
| Bus 11 | Bus 06 | 6 | -0.06072 | -0.06086 | 0.00041 | 0.00096 |
| Bus 11 | Bus 10 | 7 | 0.08737 | 0.09897 | 0.00038 | 0.001 |
| Bus 09 | Bus 10 | 8 | -0.00133 | -0.04253 | 0.00016 | 0.00034 |
| Bus 09 | Bus 14 | 9 | 0.09762 | 0.00963 | 0.00086 | 0.00182 |
| Bus 14 | Bus 13 | 10 | -0.04479 | -0.03969 | 0.00044 | 0.00089 |
| Bus 01 | Bus 02 | 11 | 1.1616 | -0.25905 | 0.07973 | 0.16707 |
| Bus 03 | Bus 02 | 12 | -0.61519 | -0.02956 | 0.01656 | 0.02585 |
| Bus 03 | Bus 04 | 13 | -0.27971 | 0.07031 | 0.00536 | -0.02002 |
| Bus 01 | Bus 05 | 14 | 1.0999 | 0.03769 | 0.05546 | 0.17848 |
| Bus 05 | Bus 04 | 15 | 0.73077 | -0.1641 | 0.00685 | 0.00905 |
| Bus 02 | Bus 04 | 16 | 0.4001 | 0.04795 | 0.00832 | -0.01245 |
| Bus 05 | Bus 06 | 17 | 0.46235 | 0.12018 | 0 | 0.04586 |
| Bus 04 | Bus 09 | 18 | 0.14137 | 0.01645 | 0 | 0.00976 |
| Bus 04 | Bus 07 | 19 | 0.23517 | -0.07686 | 0 | 0.01189 |
| Bus 08 | Bus 07 | 20 | 0 | 0.22114 | 0 | 0.00725 |
| Bus 07 | Bus 09 | 21 | 0.23517 | 0.12514 | 0 | 0.00702 |

LINE FLOWS

| From Bus | To Bus | Line | P Flow [p.u.] | Q Flow [p.u.] | P Loss [p.u.] | Q Loss [p.u.] |
|---|---|---|---|---|---|---|
| Bus 05 | Bus 02 | 1 | -0.22083 | -0.11207 | 0.00302 | -0.02512 |
| Bus 12 | Bus 06 | 2 | -0.07233 | -0.02745 | 0.00076 | 0.00158 |
| Bus 13 | Bus 12 | 3 | -0.0143 | -0.01217 | 8e-05 | 7e-05 |
| Bus 13 | Bus 06 | 4 | -0.15918 | -0.08351 | 0.00222 | 0.00438 |
| Bus 11 | Bus 06 | 5 | -0.0599 | -0.05521 | 0.00043 | 0.0009 |
| Bus 06 | Bus 11 | 6 | 0.06113 | 0.06182 | 0.00041 | 0.00096 |
| Bus 10 | Bus 11 | 7 | -0.087 | -0.09797 | 0.00038 | 0.001 |
| Bus 10 | Bus 09 | 8 | 0.0015 | 0.04287 | 0.00016 | 0.00034 |
| Bus 14 | Bus 09 | 9 | -0.09676 | -0.00781 | 0.00086 | 0.00182 |
| Bus 13 | Bus 14 | 10 | 0.04523 | 0.04058 | 0.00044 | 0.00089 |
| Bus 02 | Bus 01 | 11 | -1.0819 | 0.42612 | 0.07973 | 0.16707 |
| Bus 02 | Bus 03 | 12 | 0.63176 | 0.0554 | 0.01656 | 0.02585 |
| Bus 04 | Bus 03 | 13 | 0.28506 | -0.09034 | 0.00536 | -0.02002 |
| Bus 05 | Bus 01 | 14 | -1.0445 | 0.14079 | 0.05546 | 0.17848 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Bus 04 | Bus 05 | 15 | -0.72392 | 0.17315 | 0.00685 | 0.00905 |
| Bus 04 | Bus 02 | 16 | -0.39178 | -0.06039 | 0.00832 | -0.01245 |
| Bus 06 | Bus 05 | 17 | -0.46235 | -0.07432 | 0 | 0.04586 |
| Bus 09 | Bus 04 | 18 | -0.14137 | -0.00669 | 0 | 0.00976 |
| Bus 07 | Bus 04 | 19 | -0.23517 | 0.08875 | 0 | 0.01189 |
| Bus 07 | Bus 08 | 20 | 0 | -0.21389 | 0 | 0.00725 |
| Bus 09 | Bus 07 | 21 | -0.23517 | -0.11812 | 0 | 0.00702 |

GLOBAL SUMMARY REPORT

TOTAL GENERATION

REAL POWER [p.u.] 2.6415
REACTIVE POWER [p.u.] 1.1899

TOTAL LOAD

REAL POWER [p.u.] 2.4605
REACTIVE POWER [p.u.] 0.7733

TOTAL LOSSES

REAL POWER [p.u.] 0.18103
REACTIVE POWER [p.u.] 0.41659

LIMIT VIOLATION STATISTICS
ALL VOLTAGES WITHIN LIMITS.
ALL REACTIVE POWER WITHIN LIMITS.
ALL CURRENT FLOWS WITHIN LIMITS.
ALL REAL POWER FLOWS WITHIN LIMITS.
ALL APPARENT POWER FLOWS WITHIN LIMITS.

# Appendix C power flow Q1(c), Bus 12 and Bus 14

## *Bus 12*

NETWORK STATISTICS

Buses: 14
Lines: 21
Generators: 6
Loads: 11

SOLUTION STATISTICS

Number of Iterations: 9

Maximum P mismatch [p.u.] 0
Maximum Q mismatch [p.u.] 0
Power rate [MVA] 100

POWER FLOW RESULTS

| Bus | V [p.u.] | phase [rad] | P gen [p.u.] | Q gen [p.u.] | P load [p.u.] | Q load [p.u.] |
|---|---|---|---|---|---|---|
| Bus 01 | 1.06 | 0 | 3.6223 | 0.94607 | 0 | 0 |
| Bus 02 | 0.93058 | -0.29467 | 0.38 | 0.5 | 0.29892 | 0.17494 |
| Bus 03 | 0.88256 | -0.49552 | 0 | 0.4 | 1.2478 | 0.25168 |
| Bus 04 | 0.89389 | -0.40561 | 0 | 0 | 0.64953 | 0.05435 |
| Bus 05 | 0.90642 | -0.35245 | 0 | 0 | 0.10469 | 0.02204 |
| Bus 06 | 0.97165 | -0.52954 | 0 | 0.24 | 0.15428 | 0.10331 |
| Bus 07 | 0.93186 | -0.4887 | 0 | 0 | 0 | 0 |
| Bus 08 | 0.97521 | -0.4887 | 0 | 0.24 | 0 | 0 |
| Bus 09 | 0.91671 | -0.53225 | 0 | 0 | 0.40636 | 0.22867 |
| Bus 10 | 0.93568 | -0.5413 | 0 | 0 | 0.12398 | 0.07989 |
| Bus 11 | 0.95031 | -0.53667 | 0 | 0 | 0.04821 | 0.0248 |
| Bus 12 | 1 | -0.59443 | 0 | 0.35815 | 0.08403 | 0.02204 |
| Bus 13 | 0.94962 | -0.56231 | 0 | 0 | 0.18596 | 0.07989 |
| Bus 14 | 0.90952 | -0.56897 | 0 | 0 | 0.20525 | 0.06888 |

Maximum reactive power at bus <Bus 02>
Minimum voltage limit violation at bus <Bus 03> [V_min = 0.9]
Maximum reactive power at bus <Bus 03>
Minimum voltage limit violation at bus <Bus 04> [V_min = 0.9]
Maximum reactive power at bus <Bus 06>
Maximum reactive power at bus <Bus 08>

LINE FLOWS

| From Bus | To Bus | Line | P Flow [p.u.] | Q Flow [p.u.] | P Loss [p.u.] | Q Loss [p.u.] |
|---|---|---|---|---|---|---|
| Bus 02 | Bus 05 | 1 | 0.30906 | 0.02939 | 0.00609 | -0.00867 |
| Bus 06 | Bus 12 | 2 | 0.1414 | -0.15539 | 0.00655 | 0.01363 |
| Bus 12 | Bus 13 | 3 | 0.05082 | 0.16709 | 0.00768 | 0.00695 |
| Bus 06 | Bus 13 | 4 | 0.22135 | 0.03508 | 0.00401 | 0.0079 |
| Bus 06 | Bus 11 | 5 | 0.08883 | 0.09492 | 0.00129 | 0.00271 |
| Bus 11 | Bus 06 | 6 | -0.08824 | -0.10109 | 0.00124 | 0.00291 |
| Bus 11 | Bus 10 | 7 | 0.12757 | 0.16851 | 0.0012 | 0.00318 |
| Bus 09 | Bus 10 | 8 | -0.00159 | -0.08373 | 0.00081 | 0.00171 |
| Bus 09 | Bus 14 | 9 | 0.1354 | -0.02885 | 0.0022 | 0.00469 |
| Bus 14 | Bus 13 | 10 | -0.07205 | -0.10242 | 0.00246 | 0.00501 |
| Bus 01 | Bus 02 | 11 | 1.9042 | 0.34149 | 0.21278 | 0.51038 |
| Bus 03 | Bus 02 | 12 | -0.85564 | 0.04937 | 0.0422 | 0.14359 |
| Bus 03 | Bus 04 | 13 | -0.39217 | 0.09895 | 0.01359 | 0.00875 |
| Bus 01 | Bus 05 | 14 | 1.7181 | 0.60458 | 0.15303 | 0.58625 |
| Bus 05 | Bus 04 | 15 | 1.068 | -0.03105 | 0.01762 | 0.04572 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Bus 02 | Bus 04 | 16 | 0.56561 | 0.03256 | 0.02054 | 0.03275 |
| Bus 05 | Bus 06 | 17 | 0.69535 | 0.0654 | 0 | 0.12347 |
| Bus 04 | Bus 09 | 18 | 0.20214 | 0.0229 | 0 | 0.0257 |
| Bus 04 | Bus 07 | 19 | 0.33803 | -0.06402 | 0 | 0.02963 |
| Bus 08 | Bus 07 | 20 | 0 | 0.24 | 0 | 0.01067 |
| Bus 07 | Bus 09 | 21 | 0.33803 | 0.13569 | 0 | 0.01681 |

LINE FLOWS

| From Bus | To Bus | Line | P Flow | Q Flow | P Loss | Q Loss |
|---|---|---|---|---|---|---|
| | | [p.u.] | [p.u.] | [p.u.] | [p.u.] | |
| Bus 05 | Bus 02 | 1 | -0.30298 | -0.03807 | 0.00609 | -0.00867 |
| Bus 12 | Bus 06 | 2 | -0.13485 | 0.16902 | 0.00655 | 0.01363 |
| Bus 13 | Bus 12 | 3 | -0.04314 | -0.16014 | 0.00768 | 0.00695 |
| Bus 13 | Bus 06 | 4 | -0.21734 | -0.02718 | 0.00401 | 0.0079 |
| Bus 11 | Bus 06 | 5 | -0.08754 | -0.09222 | 0.00129 | 0.00271 |
| Bus 06 | Bus 11 | 6 | 0.08948 | 0.104 | 0.00124 | 0.00291 |
| Bus 10 | Bus 11 | 7 | -0.12637 | -0.16534 | 0.0012 | 0.00318 |
| Bus 10 | Bus 09 | 8 | 0.0024 | 0.08544 | 0.00081 | 0.00171 |
| Bus 14 | Bus 09 | 9 | -0.13319 | 0.03354 | 0.0022 | 0.00469 |
| Bus 13 | Bus 14 | 10 | 0.07452 | 0.10743 | 0.00246 | 0.00501 |
| Bus 02 | Bus 01 | 11 | -1.6914 | 0.16889 | 0.21278 | 0.51038 |
| Bus 02 | Bus 03 | 12 | 0.89784 | 0.09422 | 0.0422 | 0.14359 |
| Bus 04 | Bus 03 | 13 | 0.40576 | -0.0902 | 0.01359 | 0.00875 |
| Bus 05 | Bus 01 | 14 | -1.5651 | -0.01833 | 0.15303 | 0.58625 |
| Bus 04 | Bus 05 | 15 | -1.0504 | 0.07677 | 0.01762 | 0.04572 |
| Bus 04 | Bus 02 | 16 | -0.54507 | 0.00019 | 0.02054 | 0.03275 |
| Bus 06 | Bus 05 | 17 | -0.69535 | 0.05807 | 0 | 0.12347 |
| Bus 09 | Bus 04 | 18 | -0.20214 | 0.00279 | 0 | 0.0257 |
| Bus 07 | Bus 04 | 19 | -0.33803 | 0.09365 | 0 | 0.02963 |
| Bus 07 | Bus 08 | 20 | 0 | -0.22933 | 0 | 0.01067 |
| Bus 09 | Bus 07 | 21 | -0.33803 | -0.11888 | 0 | 0.01681 |

GLOBAL SUMMARY REPORT

TOTAL GENERATION

REAL POWER [p.u.] 4.0023
REACTIVE POWER [p.u.] 2.6842

TOTAL LOAD

REAL POWER [p.u.] 3.509
REACTIVE POWER [p.u.] 1.1105

TOTAL LOSSES

REAL POWER [p.u.] 0.49329
REACTIVE POWER [p.u.] 1.5737

LIMIT VIOLATION STATISTICS
# OF VOLTAGE LIMIT VIOLATIONS: 2
ALL REACTIVE POWERS WITHIN LIMITS (4 BINDING).
ALL CURRENT FLOWS WITHIN LIMITS.
ALL REAL POWER FLOWS WITHIN LIMITS.
ALL APPARENT POWER FLOWS WITHIN LIMITS.

## Bus 14

NETWORK STATISTICS

| | |
|---|---|
| Buses: | 14 |
| Lines: | 21 |
| Generators: | 6 |
| Loads: | 11 |

SOLUTION STATISTICS

| | |
|---|---|
| Number of Iterations: | 9 |
| Maximum P mismatch [p.u.] | 0 |
| Maximum Q mismatch [p.u.] | 0 |
| Power rate [MVA] | 100 |

POWER FLOW RESULTS

| Bus | V [p.u.] | phase [rad] | P gen [p.u.] | Q gen [p.u.] | P load [p.u.] | Q load [p.u.] |
|---|---|---|---|---|---|---|
| Bus 01 | 1.06 | 0 | 3.6596 | 0.82442 | 0 | 0 |
| Bus 02 | 0.94011 | -0.29868 | 0.38 | 0.5 | 0.29892 | 0.17494 |
| Bus 03 | 0.89415 | -0.49913 | 0 | 0.4 | 1.2808 | 0.25833 |
| Bus 04 | 0.9099 | -0.40994 | 0 | 0 | 0.65844 | 0.0551 |
| Bus 05 | 0.91957 | -0.35509 | 0 | 0 | 0.10469 | 0.02204 |
| Bus 06 | 0.98454 | -0.51916 | 0 | 0.24 | 0.15428 | 0.10331 |
| Bus 07 | 0.96627 | -0.49228 | 0 | 0 | 0 | 0 |
| Bus 08 | 1.0082 | -0.49228 | 0 | 0.24 | 0 | 0 |
| Bus 09 | 0.96148 | -0.53416 | 0 | 0 | 0.40636 | 0.22867 |
| Bus 10 | 0.96173 | -0.53488 | 0 | 0 | 0.12398 | 0.0799 |
| Bus 11 | 0.97029 | -0.52862 | 0 | 0 | 0.04821 | 0.0248 |
| Bus 12 | 0.96957 | -0.54896 | 0 | 0 | 0.08403 | 0.02204 |
| Bus 13 | 0.97086 | -0.55794 | 0 | 0 | 0.18596 | 0.0799 |
| Bus 14 | 1 | -0.5924 | 0 | 0.4743 | 0.20525 | 0.06887 |

Maximum reactive power at bus <Bus 02>
Minimum voltage limit violation at bus <Bus 03> [V_min = 0.9]
Maximum reactive power at bus <Bus 03>
Maximum reactive power at bus <Bus 06>
Maximum reactive power at bus <Bus 08>

LINE FLOWS

| From Bus | To Bus | Line | P Flow [p.u.] | Q Flow [p.u.] | P Loss [p.u.] | Q Loss [p.u.] |
|---|---|---|---|---|---|---|
| Bus 02 | Bus 05 | 1 | 0.30355 | 0.01153 | 0.00568 | -0.01058 |
| Bus 06 | Bus 12 | 2 | 0.09951 | 0.00416 | 0.00143 | 0.00298 |
| Bus 12 | Bus 13 | 3 | 0.01405 | -0.02086 | 0.00017 | 0.00015 |
| Bus 06 | Bus 13 | 4 | 0.23692 | -0.02483 | 0.00441 | 0.00869 |
| Bus 06 | Bus 11 | 5 | 0.08485 | 0.05252 | 0.00074 | 0.00155 |
| Bus 11 | Bus 06 | 6 | -0.08644 | -0.05744 | 0.00071 | 0.00167 |
| Bus 11 | Bus 10 | 7 | 0.12234 | 0.08361 | 0.00056 | 0.0015 |
| Bus 09 | Bus 10 | 8 | 0.0022 | -0.00222 | 0 | 0 |
| Bus 09 | Bus 14 | 9 | 0.15671 | -0.24598 | 0.00889 | 0.01891 |
| Bus 14 | Bus 13 | 10 | -0.05742 | 0.14054 | 0.00299 | 0.0061 |
| Bus 01 | Bus 02 | 11 | 1.9225 | 0.28199 | 0.21451 | 0.5145 |
| Bus 03 | Bus 02 | 12 | -0.87147 | 0.0612 | 0.04274 | 0.14505 |
| Bus 03 | Bus 04 | 13 | -0.40932 | 0.08047 | 0.01404 | 0.00908 |
| Bus 01 | Bus 05 | 14 | 1.7372 | 0.54242 | 0.15263 | 0.58406 |
| Bus 05 | Bus 04 | 15 | 1.115 | -0.10482 | 0.01879 | 0.04911 |
| Bus 02 | Bus 04 | 16 | 0.57127 | -0.00283 | 0.02039 | 0.03147 |
| Bus 05 | Bus 06 | 17 | 0.66272 | 0.06327 | 0 | 0.109 |
| Bus 04 | Bus 09 | 18 | 0.21171 | -0.02677 | 0 | 0.02729 |
| Bus 04 | Bus 07 | 19 | 0.35357 | -0.14519 | 0 | 0.03529 |
| Bus 08 | Bus 07 | 20 | 0 | 0.24 | 0 | 0.00998 |
| Bus 07 | Bus 09 | 21 | 0.35357 | 0.04954 | 0 | 0.01502 |

LINE FLOWS

| From Bus | To Bus | Line | P Flow [p.u.] | Q Flow [p.u.] | P Loss [p.u.] | Q Loss [p.u.] |
|---|---|---|---|---|---|---|
| Bus 05 | Bus 02 | 1 | -0.29787 | -0.02211 | 0.00568 | -0.01058 |
| Bus 12 | Bus 06 | 2 | -0.09808 | -0.00118 | 0.00143 | 0.00298 |
| Bus 13 | Bus 12 | 3 | -0.01388 | 0.02102 | 0.00017 | 0.00015 |
| Bus 13 | Bus 06 | 4 | -0.2325 | 0.03353 | 0.00441 | 0.00869 |
| Bus 11 | Bus 06 | 5 | -0.08411 | -0.05096 | 0.00074 | 0.00155 |
| Bus 06 | Bus 11 | 6 | 0.08715 | 0.05911 | 0.00071 | 0.00167 |
| Bus 10 | Bus 11 | 7 | -0.12178 | -0.08211 | 0.00056 | 0.0015 |
| Bus 10 | Bus 09 | 8 | -0.0022 | 0.00222 | 0 | 0 |
| Bus 14 | Bus 09 | 9 | -0.14782 | 0.26489 | 0.00889 | 0.01891 |
| Bus 13 | Bus 14 | 10 | 0.06042 | -0.13444 | 0.00299 | 0.0061 |
| Bus 02 | Bus 01 | 11 | -1.708 | 0.2325 | 0.21451 | 0.5145 |
| Bus 02 | Bus 03 | 12 | 0.91421 | 0.08386 | 0.04274 | 0.14505 |
| Bus 04 | Bus 03 | 13 | 0.42336 | -0.07138 | 0.01404 | 0.00908 |
| Bus 05 | Bus 01 | 14 | -1.5845 | 0.04163 | 0.15263 | 0.58406 |
| Bus 04 | Bus 05 | 15 | -1.0962 | 0.15393 | 0.01879 | 0.04911 |
| Bus 04 | Bus 02 | 16 | -0.55087 | 0.03431 | 0.02039 | 0.03147 |
| Bus 06 | Bus 05 | 17 | -0.66272 | 0.04573 | 0 | 0.109 |
| Bus 09 | Bus 04 | 18 | -0.21171 | 0.05405 | 0 | 0.02729 |
| Bus 07 | Bus 04 | 19 | -0.35357 | 0.18048 | 0 | 0.03529 |
| Bus 07 | Bus 08 | 20 | 0 | -0.23002 | 0 | 0.00998 |
| Bus 09 | Bus 07 | 21 | -0.35357 | -0.03452 | 0 | 0.01502 |

GLOBAL SUMMARY REPORT

TOTAL GENERATION

REAL POWER [p.u.] 4.0396
REACTIVE POWER [p.u.] 2.6787

TOTAL LOAD

REAL POWER [p.u.] 3.5509
REACTIVE POWER [p.u.] 1.1179

TOTAL LOSSES

REAL POWER [p.u.] 0.48871
REACTIVE POWER [p.u.] 1.5608

LIMIT VIOLATION STATISTICS
# OF VOLTAGE LIMIT VIOLATIONS: 1
ALL REACTIVE POWERS WITHIN LIMITS (4 BINDING).
ALL CURRENT FLOWS WITHIN LIMITS.
ALL REAL POWER FLOWS WITHIN LIMITS.
ALL APPARENT POWER FLOWS WITHIN LIMITS.

## Appendix D sequence YBUS matrices for Q2

### $Y_1$ & $Y_2$

| **Bus** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | **13** | **14** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | 3.170-12.784i | -2.091+5.336i | 0.000+0.000i | 0.000+0.000i | -1.080+4.458i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **2** | -2.091+5.336i | 7.044-25.611i | -1.195+5.034i | -1.775+5.385i | -1.791+5.467i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |

| **Bus** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | **13** | **14** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **3** | 0.000+0.000i | -1.195+5.034i | 4.163-14.857i | -2.091+5.336i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **4** | 0.000+0.000i | -1.775+5.385i | -2.091+5.336i | 11.512-40.447i | -7.201+22.714i | 0.000+0.000i | 0.000+4.890i | 0.000+0.000i | 0.000+1.953i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **5** | -1.080+4.458i | -1.791+5.467i | 0.000+0.000i | -7.201+22.714i | 10.142-37.417i | 0.000+4.482i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **6** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+4.482i | 9.198-28.104i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -5.047+11.180i | -1.339+2.786i | -2.718+5.353i | 0.000+0.000i |
| **7** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+4.890i | 0.000+0.000i | 0.000+0.000i | 0.000-19.549i | 0.000+5.677i | 0.000+9.090i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **8** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+5.677i | 0.000-14.010i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **9** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+1.953i | 0.000+0.000i | 0.000+0.000i | 0.000+9.090i | 0.000+0.000i | 4.008-19.100i | -1.874+3.986i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -1.874+3.986i |
| **10** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -1.874+3.986i | 7.086-17.675i | -5.134+13.639i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **11** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -5.047+11.180i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -5.134+13.639i | 10.211-24.834i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **12** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -1.339+2.786i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 3.575-4.775i | -2.183+1.975i | 0.000+0.000i |
| **13** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -2.718+5.353i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -2.183+1.975i | 6.516-10.425i | -1.496+3.046i |

| Bus | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **14** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -1.874+3.986i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -1.496+3.046i | 3.504-7.077i |

$Y_0$

| Bus | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | 1.057-3.146i | -0.697+1.779i | 0.000+0.000i | 0.000+0.000i | -0.360+1.486i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **2** | -0.697+1.779i | 2.477-6.974i | -0.398+1.678i | -0.592+1.795i | -0.597+1.823i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **3** | 0.000+0.000i | -0.398+1.678i | 1.972-3.522i | -0.697+1.779i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **4** | 0.000+0.000i | -0.592+1.795i | -0.697+1.779i | 4.133-18.080i | -2.398+7.574i | 0.000+0.000i | 0.000+4.890i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **5** | -0.360+1.486i | -0.597+1.823i | 0.000+0.000i | -2.398+7.574i | 3.426-12.895i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **6** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 3.128-6.504i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -1.682+3.727i | -0.446+0.929i | -0.906+1.785i | 0.000+0.000i |
| **7** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+4.890i | 0.000+0.000i | 0.000+0.000i | 0.000-19.552i | 0.000+5.679i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **8** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+5.679i | 0.000-14.012i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |

| Bus | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **9** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 1.510-2.804i | -0.625+1.329i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -0.625+1.329i |
| **10** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -0.625+1.329i | 2.415-5.927i | -1.712+4.548i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **11** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -1.682+3.727i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -1.712+4.548i | 3.424-8.291i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i |
| **12** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -0.446+0.929i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 1.227-1.601i | -0.728+0.659i | 0.000+0.000i |
| **13** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -0.906+1.785i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -0.728+0.659i | 2.251-3.509i | -0.499+1.015i |
| **14** | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -0.625+1.329i | 0.000+0.000i | 0.000+0.000i | 0.000+0.000i | -0.499+1.015i | 1.258-2.389i |

## Appendix E fault results for Q2

--- SHORT CIRCUIT REPORT (BUS 6) ---

THEVENIN EQUIVALENT IMPEDANCES (p.u.):
Z1: 0.013710 + j(0.110986)
Z2: 0.013710 + j(0.110986)
Z0: 1.144915 + j(0.732181)

TOTAL FAULT CURRENTS (kA):
3-Phase: 39.873
1-Phase: 8.850
L-L: 34.531
L-L-G: 4.793

NODAL VOLTAGES (kV Line-to-Ground)
Bus | 3-Phase (L-G) | Ph-A (L-G) | Ph-B (L-G) | Ph-C (L-G)
-------------------------------------------------------

| | | | | |
|---|---|---|---|---|
| 01 | 32.0083 | 40.8040 | 41.5575 | 42.1923 |
| 02 | 30.0016 | 39.8127 | 40.3641 | 41.2374 |
| 03 | 30.7044 | 39.2295 | 39.5589 | 40.4093 |
| 04 | 24.8619 | 38.3860 | 39.1505 | 40.3544 |
| 05 | 23.4605 | 38.1936 | 39.1476 | 40.3971 |
| 06 | 0.0000 | 0.0000 | 12.8857 | 14.2978 |
| 07 | 5.0683 | 8.0361 | 8.1366 | 8.4407 |
| 08 | 9.5776 | 11.1440 | 11.1995 | 11.3592 |
| 09 | 3.6899 | 2.3925 | 10.9593 | 13.2122 |
| 10 | 1.4568 | 1.0801 | 11.8813 | 13.6868 |
| 11 | 0.7906 | 0.6041 | 12.3099 | 13.9285 |
| 12 | 0.3605 | 0.5122 | 12.2446 | 13.9332 |
| 13 | 0.7303 | 0.7744 | 11.9884 | 13.7105 |
| 14 | 2.3710 | 1.8860 | 11.0444 | 13.1872 |

# Appendix F Q3 Tripped Line (2-4) case (power flow, CPF, and PV points)

## Power Flow Tripped Case

NETWORK STATISTICS

Buses: 14
Lines: 21
Generators: 5
Loads: 11

SOLUTION STATISTICS

Number of Iterations: 8
Maximum P mismatch [p.u.] 0
Maximum Q mismatch [p.u.] 0
Power rate [MVA] 100

POWER FLOW RESULTS

| Bus | V [p.u.] | phase [rad] | P gen [p.u.] | Q gen [p.u.] | P load [p.u.] | Q load [p.u.] |
|---|---|---|---|---|---|---|
| Bus 01 | 1.06 | 0 | 2.2767 | -0.073 | 0 | 0 |
| Bus 02 | 1.0354 | -0.17108 | 0.3729 | 0.5 | 0.20615 | 0.12065 |
| Bus 03 | 1.01 | -0.30873 | 0 | 0.3811 | 0.8949 | 0.1805 |
| Bus 04 | 0.99669 | -0.27464 | 0 | 0 | 0.4541 | 0.038 |
| Bus 05 | 1.0019 | -0.23321 | 0 | 0 | 0.0722 | 0.0152 |
| Bus 06 | 1.0548 | -0.33467 | 0 | 0.24 | 0.1064 | 0.07125 |
| Bus 07 | 1.0383 | -0.31907 | 0 | 0 | 0 | 0 |
| Bus 08 | 1.0775 | -0.31907 | 0 | 0.24 | 0 | 0 |
| Bus 09 | 1.0247 | -0.34231 | 0 | 0 | 0.28025 | 0.1577 |
| Bus 10 | 1.0341 | -0.34371 | 0 | 0 | 0.0855 | 0.0551 |
| Bus 11 | 1.0424 | -0.34005 | 0 | 0 | 0.03325 | 0.0171 |
| Bus 12 | 1.0371 | -0.35081 | 0 | 0 | 0.05795 | 0.0152 |
| Bus 13 | 1.0307 | -0.35134 | 0 | 0 | 0.12825 | 0.0551 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Bus 14 | 1.0141 | -0.35941 | 0 | 0 | 0.14155 | 0.0475 |

Maximum reactive power at bus <Bus 02>
Maximum reactive power at bus <Bus 06>
Maximum reactive power at bus <Bus 08>

LINE FLOWS

| From Bus | To Bus | Line | P Flow [p.u.] | Q Flow [p.u.] | P Loss [p.u.] | Q Loss [p.u.] |
|---|---|---|---|---|---|---|
| Bus 02 | Bus 05 | 1 | 0.41792 | 0.06818 | 0.00918 | -0.00549 |
| Bus 06 | Bus 12 | 2 | 0.07436 | 0.02876 | 0.0008 | 0.00167 |
| Bus 12 | Bus 13 | 3 | 0.01561 | 0.01189 | 9e-05 | 8e-05 |
| Bus 06 | Bus 13 | 4 | 0.1665 | 0.08763 | 0.0024 | 0.00473 |
| Bus 06 | Bus 11 | 5 | 0.06563 | 0.0555 | 0.00048 | 0.001 |
| Bus 11 | Bus 06 | 6 | -0.0663 | -0.06033 | 0.00046 | 0.00108 |
| Bus 11 | Bus 10 | 7 | 0.0982 | 0.09772 | 0.00043 | 0.00113 |
| Bus 09 | Bus 10 | 8 | -0.01211 | -0.04113 | 0.00017 | 0.00036 |
| Bus 09 | Bus 14 | 9 | 0.09147 | 0.01059 | 0.00078 | 0.00166 |
| Bus 14 | Bus 13 | 10 | -0.05086 | -0.03857 | 0.00051 | 0.00105 |
| Bus 01 | Bus 02 | 11 | 1.085 | -0.18455 | 0.06826 | 0.13813 |
| Bus 03 | Bus 02 | 12 | -0.74113 | 0.0708 | 0.02441 | 0.0593 |
| Bus 03 | Bus 04 | 13 | -0.15377 | 0.1298 | 0.00282 | -0.0259 |
| Bus 01 | Bus 05 | 14 | 1.1917 | 0.11155 | 0.06574 | 0.22167 |
| Bus 05 | Bus 04 | 15 | 0.98283 | -0.16661 | 0.01253 | 0.02738 |
| Bus 02 | Bus 04 | 16 | 0 | 0 | 0 | 0 |
| Bus 05 | Bus 06 | 17 | 0.47966 | 0.11496 | 0 | 0.05041 |
| Bus 04 | Bus 09 | 18 | 0.13488 | 0.01216 | 0 | 0.00916 |
| Bus 04 | Bus 07 | 19 | 0.22473 | -0.08845 | 0 | 0.01174 |
| Bus 08 | Bus 07 | 20 | 0 | 0.24 | 0 | 0.00874 |
| Bus 07 | Bus 09 | 21 | 0.22473 | 0.13107 | 0 | 0.00691 |

LINE FLOWS

| From Bus | To Bus | Line | P Flow [p.u.] | Q Flow [p.u.] | P Loss [p.u.] | Q Loss [p.u.] |
|---|---|---|---|---|---|---|
| Bus 05 | Bus 02 | 1 | -0.40874 | -0.07367 | 0.00918 | -0.00549 |
| Bus 12 | Bus 06 | 2 | -0.07356 | -0.02709 | 0.0008 | 0.00167 |
| Bus 13 | Bus 12 | 3 | -0.01552 | -0.01181 | 9e-05 | 8e-05 |
| Bus 13 | Bus 06 | 4 | -0.1641 | -0.08291 | 0.0024 | 0.00473 |
| Bus 11 | Bus 06 | 5 | -0.06516 | -0.05449 | 0.00048 | 0.001 |
| Bus 06 | Bus 11 | 6 | 0.06676 | 0.06141 | 0.00046 | 0.00108 |
| Bus 10 | Bus 11 | 7 | -0.09778 | -0.09659 | 0.00043 | 0.00113 |
| Bus 10 | Bus 09 | 8 | 0.01228 | 0.04149 | 0.00017 | 0.00036 |
| Bus 14 | Bus 09 | 9 | -0.09069 | -0.00893 | 0.00078 | 0.00166 |
| Bus 13 | Bus 14 | 10 | 0.05138 | 0.03962 | 0.00051 | 0.00105 |
| Bus 02 | Bus 01 | 11 | -1.0167 | 0.32267 | 0.06826 | 0.13813 |
| Bus 02 | Bus 03 | 12 | 0.76554 | -0.0115 | 0.02441 | 0.0593 |
| Bus 04 | Bus 03 | 13 | 0.15658 | -0.1557 | 0.00282 | -0.0259 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Bus 05 | Bus 01 | 14 | -1.126 | 0.11012 | 0.06574 | 0.22167 |
| Bus 04 | Bus 05 | 15 | -0.97029 | 0.19399 | 0.01253 | 0.02738 |
| Bus 04 | Bus 02 | 16 | 0 | 0 | 0 | 0 |
| Bus 06 | Bus 05 | 17 | -0.47966 | -0.06455 | 0 | 0.05041 |
| Bus 09 | Bus 04 | 18 | -0.13488 | -0.003 | 0 | 0.00916 |
| Bus 07 | Bus 04 | 19 | -0.22473 | 0.1002 | 0 | 0.01174 |
| Bus 07 | Bus 08 | 20 | 0 | -0.23126 | 0 | 0.00874 |
| Bus 09 | Bus 07 | 21 | -0.22473 | -0.12416 | 0 | 0.00691 |

GLOBAL SUMMARY REPORT

TOTAL GENERATION

REAL POWER [p.u.] 2.6496
REACTIVE POWER [p.u.] 1.2881

TOTAL LOAD

REAL POWER [p.u.] 2.4605
REACTIVE POWER [p.u.] 0.7733

TOTAL LOSSES

REAL POWER [p.u.] 0.18906
REACTIVE POWER [p.u.] 0.5148

LIMIT VIOLATION STATISTICS
ALL VOLTAGES WITHIN LIMITS.
ALL REACTIVE POWERS WITHIN LIMITS (3 BINDING).
ALL CURRENT FLOWS WITHIN LIMITS.
ALL REAL POWER FLOWS WITHIN LIMITS.
ALL APPARENT POWER FLOWS WITHIN LIMITS.

## CPF Tripped Case

NETWORK STATISTICS

Buses: 14
Lines: 21
Generators: 1
Loads: 11

SOLUTION STATISTICS

Number of Iterations: 50
Maximum P mismatch [p.u.] 0
Maximum Q mismatch [p.u.] 0
Power rate [MVA] 100

POWER FLOW RESULTS

| Bus | V [p.u.] | phase [rad] | P gen [p.u.] | Q gen [p.u.] | P load [p.u.] | Q load [p.u.] |
|---|---|---|---|---|---|---|
| Bus 01 | 1.06 | 0 | 3.6197 | 2.3798 | 0 | 0 |
| Bus 02 | 0.83561 | -0.26058 | 0.59288 | 0.5 | 0.29351 | 0.17178 |
| Bus 03 | 0.72117 | -0.60273 | 0 | 0.4 | 1.2741 | 0.25699 |
| Bus 04 | 0.70512 | -0.51879 | 0 | 0 | 0.64653 | 0.0541 |
| Bus 05 | 0.73868 | -0.41371 | 0 | 0 | 0.1028 | 0.02164 |
| Bus 06 | 0.70912 | -0.70553 | 0 | 0.24 | 0.15149 | 0.10144 |
| Bus 07 | 0.7063 | -0.6578 | 0 | 0 | 0 | 0 |
| Bus 08 | 0.76179 | -0.6578 | 0 | 0.24 | 0 | 0 |
| Bus 09 | 0.66964 | -0.73636 | 0 | 0 | 0.39901 | 0.22453 |
| Bus 10 | 0.67428 | -0.73685 | 0 | 0 | 0.12173 | 0.07845 |
| Bus 11 | 0.68758 | -0.72406 | 0 | 0 | 0.04734 | 0.02435 |
| Bus 12 | 0.67186 | -0.75863 | 0 | 0 | 0.08251 | 0.02164 |
| Bus 13 | 0.65996 | -0.76216 | 0 | 0 | 0.1826 | 0.07845 |
| Bus 14 | 0.63482 | -0.79388 | 0 | 0 | 0.20153 | 0.06763 |

Minimum voltage limit violation at bus <Bus 02> [V_min = 0.9]
Minimum voltage limit violation at bus <Bus 03> [V_min = 0.9]
Minimum voltage limit violation at bus <Bus 04> [V_min = 0.95]
Minimum voltage limit violation at bus <Bus 05> [V_min = 0.95]
Minimum voltage limit violation at bus <Bus 06> [V_min = 0.9]
Minimum voltage limit violation at bus <Bus 07> [V_min = 0.8]
Minimum voltage limit violation at bus <Bus 08> [V_min = 0.8]
Minimum voltage limit violation at bus <Bus 09> [V_min = 0.95]
Minimum voltage limit violation at bus <Bus 10> [V_min = 0.95]
Minimum voltage limit violation at bus <Bus 11> [V_min = 0.95]
Minimum voltage limit violation at bus <Bus 12> [V_min = 0.95]
Minimum voltage limit violation at bus <Bus 13> [V_min = 0.95]
Minimum voltage limit violation at bus <Bus 14> [V_min = 0.95]

LINE FLOWS

| From Bus | To Bus | Line | P Flow [p.u.] | Q Flow [p.u.] | P Loss [p.u.] | Q Loss [p.u.] |
|---|---|---|---|---|---|---|
| Bus 02 | Bus 05 | 1 | 0.67273 | 0.30246 | 0.04269 | 0.11026 |
| Bus 06 | Bus 12 | 2 | 0.10672 | 0.04164 | 0.00366 | 0.00761 |
| Bus 12 | Bus 13 | 3 | 0.02056 | 0.01238 | 0.00032 | 0.00029 |
| Bus 06 | Bus 13 | 4 | 0.23862 | 0.11864 | 0.01065 | 0.02097 |
| Bus 06 | Bus 11 | 5 | 0.08818 | 0.05953 | 0.00162 | 0.0034 |
| Bus 11 | Bus 06 | 6 | -0.08879 | -0.06299 | 0.00156 | 0.00366 |
| Bus 11 | Bus 10 | 7 | 0.12801 | 0.09477 | 0.0013 | 0.00345 |
| Bus 09 | Bus 10 | 8 | -0.00494 | -0.01279 | 4e-05 | 9e-05 |
| Bus 09 | Bus 14 | 9 | 0.14241 | 0.04996 | 0.00491 | 0.01044 |
| Bus 14 | Bus 13 | 10 | -0.06403 | -0.02811 | 0.00158 | 0.00321 |
| Bus 01 | Bus 02 | 11 | 1.7773 | 0.93312 | 0.23028 | 0.5578 |
| Bus 03 | Bus 02 | 12 | -1.0746 | -0.00886 | 0.09912 | 0.39223 |
| Bus 03 | Bus 04 | 13 | -0.19955 | 0.15187 | 0.00802 | 0.00376 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Bus 01 | Bus 05 | 14 | 1.8424 | 1.4467 | 0.25417 | 1.0102 |
| Bus 05 | Bus 04 | 15 | 1.4401 | 0.23153 | 0.04948 | 0.14975 |
| Bus 02 | Bus 04 | 16 | 0 | 0 | 0 | 0 |
| Bus 05 | Bus 06 | 17 | 0.67536 | 0.37549 | 0 | 0.22758 |
| Bus 04 | Bus 09 | 18 | 0.19907 | 0.10167 | 0 | 0.04986 |
| Bus 04 | Bus 07 | 19 | 0.33741 | 0.07411 | 0 | 0.04801 |
| Bus 08 | Bus 07 | 20 | 0 | 0.24 | 0 | 0.01748 |
| Bus 07 | Bus 09 | 21 | 0.33741 | 0.24862 | 0 | 0.03874 |

LINE FLOWS

| From Bus | To Bus | Line | P Flow [p.u.] | Q Flow [p.u.] | P Loss [p.u.] | Q Loss [p.u.] |
|---|---|---|---|---|---|---|
| Bus 05 | Bus 02 | 1 | -0.63004 | -0.1922 | 0.04269 | 0.11026 |
| Bus 12 | Bus 06 | 2 | -0.10306 | -0.03403 | 0.00366 | 0.00761 |
| Bus 13 | Bus 12 | 3 | -0.02024 | -0.01209 | 0.00032 | 0.00029 |
| Bus 13 | Bus 06 | 4 | -0.22797 | -0.09767 | 0.01065 | 0.02097 |
| Bus 11 | Bus 06 | 5 | -0.08656 | -0.05613 | 0.00162 | 0.0034 |
| Bus 06 | Bus 11 | 6 | 0.09035 | 0.06665 | 0.00156 | 0.00366 |
| Bus 10 | Bus 11 | 7 | -0.12671 | -0.09133 | 0.0013 | 0.00345 |
| Bus 10 | Bus 09 | 8 | 0.00498 | 0.01288 | 4e-05 | 9e-05 |
| Bus 14 | Bus 09 | 9 | -0.1375 | -0.03952 | 0.00491 | 0.01044 |
| Bus 13 | Bus 14 | 10 | 0.06561 | 0.03132 | 0.00158 | 0.00321 |
| Bus 02 | Bus 01 | 11 | -1.5471 | -0.37532 | 0.23028 | 0.5578 |
| Bus 02 | Bus 03 | 12 | 1.1737 | 0.40109 | 0.09912 | 0.39223 |
| Bus 04 | Bus 03 | 13 | 0.20757 | -0.14811 | 0.00802 | 0.00376 |
| Bus 05 | Bus 01 | 14 | -1.5882 | -0.43646 | 0.25417 | 1.0102 |
| Bus 04 | Bus 05 | 15 | -1.3906 | -0.08178 | 0.04948 | 0.14975 |
| Bus 04 | Bus 02 | 16 | 0 | 0 | 0 | 0 |
| Bus 06 | Bus 05 | 17 | -0.67536 | -0.14791 | 0 | 0.22758 |
| Bus 09 | Bus 04 | 18 | -0.19907 | -0.05181 | 0 | 0.04986 |
| Bus 07 | Bus 04 | 19 | -0.33741 | -0.0261 | 0 | 0.04801 |
| Bus 07 | Bus 08 | 20 | 0 | -0.22252 | 0 | 0.01748 |
| Bus 09 | Bus 07 | 21 | -0.33741 | -0.20988 | 0 | 0.03874 |

GLOBAL SUMMARY REPORT

TOTAL GENERATION

REAL POWER [p.u.] 4.2126
REACTIVE POWER [p.u.] 3.7598

TOTAL LOAD

REAL POWER [p.u.] 3.5032
REACTIVE POWER [p.u.] 1.101

TOTAL LOSSES

REAL POWER [p.u.] 0.7094

REACTIVE POWER [p.u.]    2.6588

LIMIT VIOLATION STATISTICS
# OF VOLTAGE LIMIT VIOLATIONS: 13
ALL REACTIVE POWER WITHIN LIMITS.
ALL CURRENT FLOWS WITHIN LIMITS.
ALL REAL POWER FLOWS WITHIN LIMITS.
ALL APPARENT POWER FLOWS WITHIN LIMITS.

## PV for Bus 11 Data Points Tripped Case

| **Loading Parameter λ** | **Bus 11 Voltage (p.u.)** |
|---|---|
| 0.99000 | 1.04251 |
| 0.99794 | 1.04239 |
| 1.00515 | 1.04141 |
| 1.01181 | 1.04009 |
| 1.06249 | 1.01979 |
| 1.10955 | 0.99939 |
| 1.15304 | 0.97890 |
| 1.19301 | 0.95833 |
| 1.22953 | 0.93769 |
| 1.26265 | 0.91699 |
| 1.29245 | 0.89623 |
| 1.31899 | 0.87544 |
| 1.34235 | 0.85461 |
| 1.36260 | 0.83375 |
| 1.37981 | 0.81287 |
| 1.39407 | 0.79199 |
| 1.40544 | 0.77109 |
| 1.41401 | 0.75020 |
| 1.41987 | 0.72931 |
| 1.42309 | 0.70844 |
| 1.42377 | 0.68758 |
| 1.42198 | 0.66673 |
| 1.41782 | 0.64591 |
| 1.41137 | 0.62511 |
| 1.40272 | 0.60433 |
| 1.39196 | 0.58358 |
| 1.37917 | 0.56283 |

| Loading Parameter λ | Bus 11 Voltage (p.u.) |
|---|---|
| 1.36443 | 0.54210 |
| 1.34781 | 0.52136 |
| 1.32936 | 0.50060 |
| 1.30910 | 0.47980 |
| 1.28702 | 0.45893 |
| 1.26303 | 0.43797 |
| 1.23696 | 0.41690 |
| 1.20848 | 0.39575 |
| 1.17710 | 0.37463 |
| 1.14218 | 0.35393 |
| 1.10329 | 0.33460 |
| 1.06130 | 0.31862 |
| 1.02046 | 0.30905 |
| 1.01686 | 0.30856 |
| 1.01335 | 0.30814 |
| 1.00993 | 0.30780 |
| 1.00659 | 0.30754 |
| 1.00333 | 0.30734 |
| 1.00017 | 0.30722 |
| 0.99709 | 0.30716 |
| 0.99410 | 0.30716 |
| 0.99120 | 0.30722 |
| 0.99092 | 0.30723 |

# Appendix G Q3 Base case (power flow, CPF, and PV points)

## Power Flow Base Case

NETWORK STATISTICS

Buses: 14
Lines: 21
Generators: 5
Loads: 11

SOLUTION STATISTICS

Number of Iterations: 8
Maximum P mismatch [p.u.] 0
Maximum Q mismatch [p.u.] 0
Power rate [MVA] 100

POWER FLOW RESULTS

| Bus | V [p.u.] | phase [rad] | P gen [p.u.] | Q gen [p.u.] | P load [p.u.] | Q load [p.u.] |
|---|---|---|---|---|---|---|
| Bus 01 | 1.06 | 0 | 2.2697 | -0.11348 | 0 | 0 |
| Bus 02 | 1.0322 | -0.18357 | 0.37177 | 0.5 | 0.20615 | 0.12065 |
| Bus 03 | 1.01 | -0.29803 | 0 | 0.32927 | 0.8949 | 0.1805 |
| Bus 04 | 1.009 | -0.24616 | 0 | 0 | 0.4541 | 0.038 |
| Bus 05 | 1.0113 | -0.21542 | 0 | 0 | 0.0722 | 0.0152 |
| Bus 06 | 1.0656 | -0.31132 | 0 | 0.24 | 0.1064 | 0.07125 |
| Bus 07 | 1.0507 | -0.29158 | 0 | 0 | 0 | 0 |
| Bus 08 | 1.0895 | -0.29158 | 0 | 0.24 | 0 | 0 |
| Bus 09 | 1.0373 | -0.31535 | 0 | 0 | 0.28025 | 0.1577 |
| Bus 10 | 1.0456 | -0.31881 | 0 | 0 | 0.0855 | 0.0551 |
| Bus 11 | 1.0536 | -0.31587 | 0 | 0 | 0.03348 | 0.01722 |
| Bus 12 | 1.0482 | -0.32677 | 0 | 0 | 0.05795 | 0.0152 |
| Bus 13 | 1.0421 | -0.32696 | 0 | 0 | 0.12825 | 0.0551 |
| Bus 14 | 1.0263 | -0.33327 | 0 | 0 | 0.14155 | 0.0475 |

Maximum reactive power at bus <Bus 02>
Maximum reactive power at bus <Bus 06>
Maximum reactive power at bus <Bus 08>
Maximum voltage limit violation at bus <Bus 11> [V_max = 1.05]

LINE FLOWS

| From Bus | To Bus | Line | P Flow [p.u.] | Q Flow [p.u.] | P Loss [p.u.] | Q Loss [p.u.] |
|---|---|---|---|---|---|---|
| Bus 02 | Bus 05 | 1 | 0.22131 | 0.04424 | 0.00268 | -0.02554 |
| Bus 06 | Bus 12 | 2 | 0.07306 | 0.02891 | 0.00076 | 0.00159 |
| Bus 12 | Bus 13 | 3 | 0.01434 | 0.01212 | 8e-05 | 7e-05 |
| Bus 06 | Bus 13 | 4 | 0.16137 | 0.08741 | 0.00224 | 0.00441 |
| Bus 06 | Bus 11 | 5 | 0.06037 | 0.05562 | 0.00043 | 0.0009 |
| Bus 11 | Bus 06 | 6 | -0.06078 | -0.06034 | 0.00041 | 0.00096 |
| Bus 11 | Bus 10 | 7 | 0.08724 | 0.09785 | 0.00037 | 0.00099 |
| Bus 09 | Bus 10 | 8 | -0.00121 | -0.04143 | 0.00015 | 0.00033 |
| Bus 09 | Bus 14 | 9 | 0.0977 | 0.01027 | 0.00087 | 0.00184 |
| Bus 14 | Bus 13 | 10 | -0.04471 | -0.03907 | 0.00043 | 0.00089 |
| Bus 01 | Bus 02 | 11 | 1.1657 | -0.18056 | 0.07848 | 0.16433 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Bus 03 | Bus 02 | 12 | -0.61788 | 0.04259 | 0.01689 | 0.02775 |
| Bus 03 | Bus 04 | 13 | -0.27702 | 0.10618 | 0.00573 | -0.01887 |
| Bus 01 | Bus 05 | 14 | 1.104 | 0.06707 | 0.05608 | 0.18133 |
| Bus 05 | Bus 04 | 15 | 0.73197 | -0.17013 | 0.00698 | 0.0096 |
| Bus 02 | Bus 04 | 16 | 0.39679 | 0.00507 | 0.00819 | -0.01217 |
| Bus 05 | Bus 06 | 17 | 0.46239 | 0.11046 | 0 | 0.04596 |
| Bus 04 | Bus 09 | 18 | 0.14133 | 0.0128 | 0 | 0.00981 |
| Bus 04 | Bus 07 | 19 | 0.23541 | -0.08825 | 0 | 0.01242 |
| Bus 08 | Bus 07 | 20 | 0 | 0.24 | 0 | 0.00855 |
| Bus 07 | Bus 09 | 21 | 0.23541 | 0.13079 | 0 | 0.00723 |

LINE FLOWS

| From Bus | To Bus | Line | P Flow [p.u.] | Q Flow [p.u.] | P Loss [p.u.] | Q Loss [p.u.] |
|---|---|---|---|---|---|---|
| Bus 05 | Bus 02 | 1 | -0.21864 | -0.06978 | 0.00268 | -0.02554 |
| Bus 12 | Bus 06 | 2 | -0.07229 | -0.02732 | 0.00076 | 0.00159 |
| Bus 13 | Bus 12 | 3 | -0.01426 | -0.01205 | 8e-05 | 7e-05 |
| Bus 13 | Bus 06 | 4 | -0.15913 | -0.08301 | 0.00224 | 0.00441 |
| Bus 11 | Bus 06 | 5 | -0.05994 | -0.05472 | 0.00043 | 0.0009 |
| Bus 06 | Bus 11 | 6 | 0.0612 | 0.06131 | 0.00041 | 0.00096 |
| Bus 10 | Bus 11 | 7 | -0.08687 | -0.09685 | 0.00037 | 0.00099 |
| Bus 10 | Bus 09 | 8 | 0.00137 | 0.04175 | 0.00015 | 0.00033 |
| Bus 14 | Bus 09 | 9 | -0.09684 | -0.00843 | 0.00087 | 0.00184 |
| Bus 13 | Bus 14 | 10 | 0.04515 | 0.03996 | 0.00043 | 0.00089 |
| Bus 02 | Bus 01 | 11 | -1.0873 | 0.34489 | 0.07848 | 0.16433 |
| Bus 02 | Bus 03 | 12 | 0.63477 | -0.01484 | 0.01689 | 0.02775 |
| Bus 04 | Bus 03 | 13 | 0.28275 | -0.12505 | 0.00573 | -0.01887 |
| Bus 05 | Bus 01 | 14 | -1.0479 | 0.11426 | 0.05608 | 0.18133 |
| Bus 04 | Bus 05 | 15 | -0.72499 | 0.17974 | 0.00698 | 0.0096 |
| Bus 04 | Bus 02 | 16 | -0.3886 | -0.01724 | 0.00819 | -0.01217 |
| Bus 06 | Bus 05 | 17 | -0.46239 | -0.0645 | 0 | 0.04596 |
| Bus 09 | Bus 04 | 18 | -0.14133 | -0.00299 | 0 | 0.00981 |
| Bus 07 | Bus 04 | 19 | -0.23541 | 0.10067 | 0 | 0.01242 |
| Bus 07 | Bus 08 | 20 | 0 | -0.23145 | 0 | 0.00855 |
| Bus 09 | Bus 07 | 21 | -0.23541 | -0.12356 | 0 | 0.00723 |

GLOBAL SUMMARY REPORT

TOTAL GENERATION

REAL POWER [p.u.] 2.6415
REACTIVE POWER [p.u.] 1.1958

TOTAL LOAD

REAL POWER [p.u.] 2.4607
REACTIVE POWER [p.u.] 0.77342

TOTAL LOSSES

REAL POWER [p.u.] 0.18077
REACTIVE POWER [p.u.] 0.42237

LIMIT VIOLATION STATISTICS
# OF VOLTAGE LIMIT VIOLATIONS: 1
ALL REACTIVE POWERS WITHIN LIMITS (3 BINDING).
ALL CURRENT FLOWS WITHIN LIMITS.
ALL REAL POWER FLOWS WITHIN LIMITS.
ALL APPARENT POWER FLOWS WITHIN LIMITS.

## CPF Base Case

NETWORK STATISTICS

Buses: 14
Lines: 21
Generators: 1
Loads: 11

SOLUTION STATISTICS

Number of Iterations: 50
Maximum P mismatch [p.u.] 0
Maximum Q mismatch [p.u.] 0
Power rate [MVA] 100

POWER FLOW RESULTS

| Bus | V [p.u.] | phase [rad] | P gen [p.u.] | Q gen [p.u.] | P load [p.u.] | Q load [p.u.] |
|---|---|---|---|---|---|---|
| Bus 01 | 1.06 | 0 | 3.8269 | 2.5336 | 0 | 0 |
| Bus 02 | 0.80095 | -0.30889 | 0.62681 | 0.5 | 0.30913 | 0.18092 |
| Bus 03 | 0.70467 | -0.62663 | 0 | 0.4 | 1.342 | 0.27067 |
| Bus 04 | 0.71789 | -0.47817 | 0 | 0 | 0.68095 | 0.05698 |
| Bus 05 | 0.74393 | -0.39825 | 0 | 0 | 0.10827 | 0.02279 |
| Bus 06 | 0.70938 | -0.69186 | 0 | 0.24 | 0.15955 | 0.10684 |
| Bus 07 | 0.71251 | -0.62758 | 0 | 0 | 0 | 0 |
| Bus 08 | 0.76759 | -0.62758 | 0 | 0.24 | 0 | 0 |
| Bus 09 | 0.67355 | -0.71303 | 0 | 0 | 0.42025 | 0.23648 |
| Bus 10 | 0.67494 | -0.72037 | 0 | 0 | 0.12821 | 0.08263 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Bus 11 | 0.68798 | -0.70905 | 0 | 0 | 0.04986 | 0.02564 |
| Bus 12 | 0.67054 | -0.74669 | 0 | 0 | 0.0869 | 0.02279 |
| Bus 13 | 0.65879 | -0.74946 | 0 | 0 | 0.19232 | 0.08263 |
| Bus 14 | 0.63482 | -0.77744 | 0 | 0 | 0.21226 | 0.07123 |

Minimum voltage limit violation at bus <Bus 02> [V_min = 0.9]
Minimum voltage limit violation at bus <Bus 03> [V_min = 0.9]
Minimum voltage limit violation at bus <Bus 04> [V_min = 0.95]
Minimum voltage limit violation at bus <Bus 05> [V_min = 0.95]
Minimum voltage limit violation at bus <Bus 06> [V_min = 0.9]
Minimum voltage limit violation at bus <Bus 07> [V_min = 0.8]
Minimum voltage limit violation at bus <Bus 08> [V_min = 0.8]
Minimum voltage limit violation at bus <Bus 09> [V_min = 0.95]
Minimum voltage limit violation at bus <Bus 10> [V_min = 0.95]
Minimum voltage limit violation at bus <Bus 11> [V_min = 0.95]
Minimum voltage limit violation at bus <Bus 12> [V_min = 0.95]
Minimum voltage limit violation at bus <Bus 13> [V_min = 0.95]
Minimum voltage limit violation at bus <Bus 14> [V_min = 0.95]

LINE FLOWS

| From Bus | To Bus | Line | P Flow | Q Flow | P Loss | Q Loss |
|---|---|---|---|---|---|---|
| | | | [p.u.] | [p.u.] | [p.u.] | [p.u.] |
| Bus 02 | Bus 05 | 1 | 0.37676 | 0.15715 | 0.01434 | 0.02448 |
| Bus 06 | Bus 12 | 2 | 0.11045 | 0.04386 | 0.00393 | 0.00818 |
| Bus 12 | Bus 13 | 3 | 0.01962 | 0.01288 | 0.00031 | 0.00028 |
| Bus 06 | Bus 13 | 4 | 0.24368 | 0.12311 | 0.01117 | 0.022 |
| Bus 06 | Bus 11 | 5 | 0.08442 | 0.06057 | 0.00155 | 0.00324 |
| Bus 11 | Bus 06 | 6 | -0.08486 | -0.0641 | 0.00149 | 0.00349 |
| Bus 11 | Bus 10 | 7 | 0.11787 | 0.09579 | 0.00118 | 0.00313 |
| Bus 09 | Bus 10 | 8 | 0.01157 | -0.00993 | 5e-05 | 0.00011 |
| Bus 09 | Bus 14 | 9 | 0.16024 | 0.05593 | 0.00613 | 0.01305 |
| Bus 14 | Bus 13 | 10 | -0.05816 | -0.02834 | 0.00135 | 0.00275 |
| Bus 01 | Bus 02 | 11 | 2.0352 | 1.1215 | 0.30829 | 0.75783 |
| Bus 03 | Bus 02 | 12 | -0.93488 | 0.00105 | 0.07858 | 0.30739 |
| Bus 03 | Bus 04 | 13 | -0.40707 | 0.12828 | 0.02363 | 0.04368 |
| Bus 01 | Bus 05 | 14 | 1.7917 | 1.4122 | 0.24117 | 0.95637 |
| Bus 05 | Bus 04 | 15 | 1.1202 | 0.16833 | 0.02943 | 0.08635 |
| Bus 02 | Bus 04 | 16 | 0.65432 | 0.21922 | 0.04142 | 0.10512 |
| Bus 05 | Bus 06 | 17 | 0.68446 | 0.39735 | 0 | 0.23537 |
| Bus 04 | Bus 09 | 18 | 0.21977 | 0.12029 | 0 | 0.06042 |
| Bus 04 | Bus 07 | 19 | 0.37229 | 0.10341 | 0 | 0.05794 |
| Bus 08 | Bus 07 | 20 | 0 | 0.24 | 0 | 0.01722 |
| Bus 07 | Bus 09 | 21 | 0.37229 | 0.26825 | 0 | 0.04563 |

LINE FLOWS

| From Bus | To Bus | Line | P Flow [p.u.] | Q Flow [p.u.] | P Loss [p.u.] | Q Loss [p.u.] |
|---|---|---|---|---|---|---|
| Bus 05 | Bus 02 | 1 | -0.36242 | -0.13267 | 0.01434 | 0.02448 |
| Bus 12 | Bus 06 | 2 | -0.10652 | -0.03568 | 0.00393 | 0.00818 |
| Bus 13 | Bus 12 | 3 | -0.01931 | -0.0126 | 0.00031 | 0.00028 |
| Bus 13 | Bus 06 | 4 | -0.23251 | -0.10111 | 0.01117 | 0.022 |
| Bus 11 | Bus 06 | 5 | -0.08287 | -0.05733 | 0.00155 | 0.00324 |
| Bus 06 | Bus 11 | 6 | 0.08635 | 0.06759 | 0.00149 | 0.00349 |
| Bus 10 | Bus 11 | 7 | -0.11669 | -0.09266 | 0.00118 | 0.00313 |
| Bus 10 | Bus 09 | 8 | -0.01152 | 0.01003 | 5e-05 | 0.00011 |
| Bus 14 | Bus 09 | 9 | -0.15411 | -0.04288 | 0.00613 | 0.01305 |
| Bus 13 | Bus 14 | 10 | 0.05951 | 0.03109 | 0.00135 | 0.00275 |
| Bus 02 | Bus 01 | 11 | -1.7269 | -0.36364 | 0.30829 | 0.75783 |
| Bus 02 | Bus 03 | 12 | 1.0135 | 0.30634 | 0.07858 | 0.30739 |
| Bus 04 | Bus 03 | 13 | 0.4307 | -0.0846 | 0.02363 | 0.04368 |
| Bus 05 | Bus 01 | 14 | -1.5505 | -0.4558 | 0.24117 | 0.95637 |
| Bus 04 | Bus 05 | 15 | -1.0908 | -0.08198 | 0.02943 | 0.08635 |
| Bus 04 | Bus 02 | 16 | -0.6129 | -0.1141 | 0.04142 | 0.10512 |
| Bus 06 | Bus 05 | 17 | -0.68446 | -0.16197 | 0 | 0.23537 |
| Bus 09 | Bus 04 | 18 | -0.21977 | -0.05987 | 0 | 0.06042 |
| Bus 07 | Bus 04 | 19 | -0.37229 | -0.04547 | 0 | 0.05794 |
| Bus 07 | Bus 08 | 20 | 0 | -0.22278 | 0 | 0.01722 |
| Bus 09 | Bus 07 | 21 | -0.37229 | -0.22262 | 0 | 0.04563 |

GLOBAL SUMMARY REPORT

TOTAL GENERATION

REAL POWER [p.u.] 4.4537
REACTIVE POWER [p.u.] 3.9136

TOTAL LOAD

REAL POWER [p.u.] 3.6897
REACTIVE POWER [p.u.] 1.1596

TOTAL LOSSES

REAL POWER [p.u.] 0.76402
REACTIVE POWER [p.u.] 2.754

LIMIT VIOLATION STATISTICS
# OF VOLTAGE LIMIT VIOLATIONS: 13

ALL REACTIVE POWER WITHIN LIMITS.
ALL CURRENT FLOWS WITHIN LIMITS.
ALL REAL POWER FLOWS WITHIN LIMITS.
ALL APPARENT POWER FLOWS WITHIN LIMITS.

## PV for Bus 11 Data Points Base Case

| **Loading Parameter λ** | **Bus 11 Voltage (p.u.)** |
|---|---|
| 0.99000 | 1.05375 |
| 0.99861 | 1.05363 |
| 1.00641 | 1.05257 |
| 1.01400 | 1.05128 |
| 1.02157 | 1.04999 |
| 1.02911 | 1.04870 |
| 1.03662 | 1.04741 |
| 1.04411 | 1.04612 |
| 1.04986 | 1.04409 |
| 1.10485 | 1.02360 |
| 1.15593 | 1.00300 |
| 1.20318 | 0.98231 |
| 1.24663 | 0.96154 |
| 1.28637 | 0.94070 |
| 1.32245 | 0.91979 |
| 1.35494 | 0.89883 |
| 1.38392 | 0.87783 |
| 1.40945 | 0.85679 |
| 1.43163 | 0.83573 |
| 1.45052 | 0.81464 |
| 1.46621 | 0.79353 |
| 1.47877 | 0.77242 |
| 1.48831 | 0.75131 |
| 1.49489 | 0.73019 |
| 1.49861 | 0.70908 |
| 1.49955 | 0.68798 |
| 1.49782 | 0.66689 |
| 1.49350 | 0.64582 |

| **Loading Parameter λ** | **Bus 11 Voltage (p.u.)** |
|---|---|
| 1.48670 | 0.62476 |
| 1.47750 | 0.60371 |
| 1.46600 | 0.58267 |
| 1.45232 | 0.56164 |
| 1.43654 | 0.54062 |
| 1.41875 | 0.51958 |
| 1.39904 | 0.49851 |
| 1.37746 | 0.47741 |
| 1.35402 | 0.45625 |
| 1.32866 | 0.43503 |
| 1.30119 | 0.41375 |
| 1.27128 | 0.39253 |
| 1.23835 | 0.37159 |
| 1.20172 | 0.35157 |
| 1.16107 | 0.33373 |
| 1.11784 | 0.32044 |
| 1.07758 | 0.31466 |
| 1.04650 | 0.31647 |
| 1.02322 | 0.32283 |
| 1.00376 | 0.33150 |
| 1.00191 | 0.33244 |
| 1.00008 | 0.33340 |